Testis, Penis & Urethra

Condensed rapid revision · facts already answered in Papers 1–4 point to that answer

highlighted number to memorise · **KEY POINT** and **ALGORITHM** boxes carry the reasoning · MCQ exam trivia · † added beyond the source notes · P2 topic = covered in full in that Paper's answer book

Chapter	Topic
Ch 78	Anatomy of the Retroperitoneum
Ch 79	Neoplasms of the Testis
Ch 80	Surgery of Testicular Tumours
Ch 81	Minimally Invasive RPLND
Ch 82	Tumours of the Penis
Ch 84	Inguinal Node Dissection
Ch 83	Tumours of the Urethra
Ch 85	Benign Disorders of the Penis & Urethra
Ch 86	Urethral Stricture Disease
Ch 87	Surgery of the Scrotum & Seminal Vesicles
★	Putting it together

78 Surgical, Radiographic and Endoscopic Anatomy of the Retroperitoneum

1 · Definition, boundaries and organs

- P1 Describe in brief the anatomy of the retroperitoneum… retroperitoneal laparoscopic nephrectomy — *boundaries, primary vs secondary organs, flank layers, lumbodorsal fascia at 12th rib tip, three spaces, lateroconal fascia, perirenal cone open below, fat colour*
- Psoas/quadratus lumborum: boundary or content — debated. Primarily retroperitoneal also includes **anal canal**; secondarily retroperitoneal excludes **D1** and pancreatic tail; primary organs in perirenal space, secondary in anterior pararenal.

> **KEY POINT**
> Secondarily retroperitoneal organs lie **in front of** the kidney, so colon, duodenum and pancreas are always in the surgeon's path to the kidney.

2 · Posterior abdominal wall and surface landmarks

- Upper pole: left 11th rib, right 12th rib (right lower). Hilum **L1**; lower pole L3–L4.
- External oblique inferomedial ("hands in pockets"); internal oblique superomedial.
- **Psoas major** T12–L5 → lesser trochanter (hip flexion, with iliacus). **Psoas minor** inconstant; inserts at pelvic brim. **Quadratus lumborum** posterolateral to psoas; stabilises 12th rib.
- Lower rib fracture → suspect retroperitoneal injury. **Costovertebral (lumbocostal) ligament** tethers 12th rib to L1–L2 transverse processes; dividing it frees the rib in posterior approaches.
- **Intercostal bundle**: groove of rib above; vein–artery–nerve top-down — nerve lowest, least protected. Supracostal percutaneous access may cross pleura → more complications than subcostal.
- **Dorsal lumbotomy**: vertical incision through lumbodorsal fascia lateral to erector spinae and quadratus — **no muscle cut**.

3 · Retroperitoneal fasciae and spaces

Three fetal strata → three fasciae → three spaces: outer (transversalis; posterior pararenal), intermediate (Gerota; perirenal, genitourinary), inner (fusion fascia with dorsal peritoneum; anterior pararenal, gastrointestinal).

- **Posterior pararenal**: medial fusion of posterior Gerota lamina and transversalis **must be incised to reach the hilum** retroperitoneoscopically.
- Toldt (anterior) lamina thinner; Zuckerkandl (posterior) thicker, seen more on imaging. MCQ Gerota described only the anterior, yet his name covers both.
- Perirenal cone may communicate across midline below hilum; urinoma, haematoma (ruptured aneurysm), perinephric abscess contained laterally but **track caudally**. Fat colour change guides plane in colon mobilisation.
- **Anterior pararenal**: contains colon, pancreas, D2–D3; entered from the **white line of Toldt**.

4 · Gastrointestinal relations (operative hazards)

- **Right**: pancreatic head anterior to IVC; D2 on right hilum → **Kocher manoeuvre**. D3 crosses between SMA (front) and aorta.
- **Left**: pancreatic tail, spleen, stomach near upper pole; splenic vessels on upper pancreatic border at risk lifting pancreas off renal fascia.
- Divide hepatocolic/splenocolic ligaments sharply; avoid over-retraction. Close small mesenteric defects (higher internal-hernia risk).

5 · Great vessels

Aorta and branches

Great vessel compartment T12 to L4/5. Aorta enters at **T12**, left of IVC; bifurcates at **L4**.

Branch	Type	Level	Supplies
Inferior phrenic	Paired lateral	First, below hiatus	Diaphragm; superior adrenal
Celiac	Anterior	T12	Foregut
Middle adrenal	Paired lateral	SMA level	Adrenal
SMA	Anterior	L1–L2	Midgut
Renal	Paired lateral	L1–L2, below SMA	Kidney; inferior adrenal
Gonadal	Paired **anterolateral**	Below renals	Testis/ovary
Lumbar	Paired posterior	Four pairs, L1–L4	Wall, spine, cord
IMA	Anterior	L3–L4	Hindgut

- Adrenal arteries: superior constant; middle, inferior variable. Supernumerary renal arteries ≥ ¼ → *Ch 84*.
- **Testicular artery** anterior to psoas, genitofemoral nerve, ureter (and IVC on right) to deep ring.
- **Ureter**: proximal renal; middle aorta, common iliac, gonadal; distal internal iliac. Abdominal ureter supplied **medially**, pelvic **laterally**.
- IMA ligation safe: marginal artery of **Drummond** (colon); superior ↔ middle/inferior rectal (rectum). **SMA not safe** — severe bowel ischaemia (risk in left-sided surgery).

IVC and veins

- IVC forms below/right of aortic bifurcation; passes diaphragm at **T8**. Veins vary more than arteries.
- **Left renal vein**: crosses anterior to aorta below SMA; gonadal (right-angle entry), adrenal (± inferior phrenic), ≥ 1 **lumbar vein** (avulsion risk). **Right**: short, no tributaries; duplicated in one-sixth. Enter IVC at L1.
- **Right gonadal vein** → anterolateral IVC (right renal vein in up to 10%); **right adrenal vein** short, posterolateral IVC.
- Left varicocele predominance: longer vein, perpendicular entry. **New right varicocele** → image for retroperitoneal mass (e.g. RCC with venous thrombus).
- Lumbar veins → ascending lumbar (behind psoas) → azygos/hemiazygos.
- Right ureter: dissect **lateral** to gonadal vein (medial risks IVC, gonadal avulsion).
- **IMV** in left mesocolon; joins SMV in 40%; at reoperation mistaken for gonadal vein.

Variants (essential)

- P1 P3 IVC anomalous development and retrocaval ureter; Retrocaval ureter — *renal collar embryology, circumaortic/retroaortic vein, retrocaval ureter types and repair*
- † Retroaortic left renal vein ≈ 2–3%: injured dividing lumbars/rolling aorta in RPLND. Circumaortic ≈ 5–7%: posterior limb mimics a lumbar vein — check pre-operative CT. Retrocaval ureter: **venous** anomaly; "reverse-J" on pyelogram; dismembered pyeloplasty anterior to IVC.

6 · Lymphatics

- P1 Lymphatic drainage of testis and scrotum, RPLND templates and penile cancer implications — *right→left cephalad flow, cisterna chyli L1–L2, three nodal regions, landing zones (right interaortocaval, left para-aortic), right→left crossover only*
- Lumbar nodes = primary drainage of lateral-branch organs (kidney, adrenal, ureter, gonad).

> **KEY POINT**
> Testis is embryologically retroperitoneal — drains to where it began. Crossover one-way (right → left), so right templates include interaortocaval and some para-aortic tissue. → *Ch 80, 81*

7 · Nerves

Autonomic

- **Adrenal medulla** = modified sympathetic ganglion (preganglionic fibres synapse directly).
- **Parasympathetic**: craniosacral (III, VII, IX, X, S2–S4); ganglia in organ wall. **Sympathetic**: thoracolumbar **T1–L2**.
- **Sympathetic trunks**: medial to psoas; behind great vessels; lumbar vessels cross them at right angles — met when lumbars divided.
- Renal plexus (aorticorenal ganglion): kidney, ureter, some testicular fibres. Superior hypogastric plexus: caudal aorta to front of L5.
- **Ejaculation**: † postganglionic fibres from **T12–L3** ganglia run anterior to aorta/across IVC, converge in hypogastric plexus below IMA → **emission** and **bladder neck closure**. Extensive dissection → anejaculation or retrograde ejaculation; erection (parasympathetic) spared. → *Ch 81*

Lumbar (lumbosacral) plexus

Nerve	Roots	Course / motor	Sensory
Subcostal	T12	Below 12th rib	Skin over hip
Iliohypogastric	L1	Over quadratus; internal oblique, transversus	Gluteal, pubic
Ilioinguinal	L1	As above	Upper medial thigh, penile root, anterior scrotum
Genitofemoral	L1–L2	**On anterior psoas**; genital branch → cremaster	Anterior scrotum; upper anterior thigh
Lateral cutaneous thigh	L2–L3	Sensory only	Lateral thigh
Obturator	L2–L4	Medial to psoas → obturator canal; adductors	Medial thigh

Nerve	Roots	Course / motor	Sensory
Femoral	L2–L4	Lateral to psoas; quadriceps	Anterior thigh, medial leg

Nerve	Setting	Result
Genitofemoral	Psoas hitch, laparoscopic varicocelectomy, RPLND on psoas	Anterior scrotal/thigh numbness
Obturator	Lateral-wall TURBT; pelvic node dissection	Adductor jerk → bladder perforation
Femoral	Retractor blade on psoas; lithotomy	Loss of knee extension
Hypogastric plexus	Extensive retroperitoneal dissection	Anejaculation/retrograde ejaculation

8 · Imaging anatomy

- CT cranial → caudal: upper poles → renal vessels (left renal vein anterior to aorta) → pelves with **duodenum against right hilum** → lower poles.
- † Before RPLND/nephrectomy: accessory renal arteries, retro-/circumaortic left renal vein, nodal size by zone.

79 Neoplasms of the Testis

1 · Epidemiology and risk factors

- >95% are **germ cell tumours (GCTs)**: seminoma vs non-seminomatous GCT (NSGCT); 95% testicular, 5% extragonadal. Commonest solid malignancy in men **20–40 y**; peak 25–35.
- Cisplatin: metastatic cure 5–10% → **80–90%**; aim now **minimum toxicity without losing curability**. Deaths from platinum resistance and failure to clear residual disease.
- Bilateral **2%**, mostly metachronous; histology discordant 30–50%. Highest in Scandinavia; rising.
- **Five risk factors**: White race, cryptorchidism, family history, personal history, GCNIS; also infertility.
  - Undescended testis (UDT): RR **4–6**, 2–3 if orchidopexy before puberty; contralateral descended ≈1.7.
  - Brother RR 8–12 > father 2–4. Previous GCT: 12-fold contralateral risk, 15-y cumulative incidence only 2%.
- **GCNIS**: arrested fetal gonocyte, formed **before birth**; beside invasive GCT in 80–90%; invasion **50% at 5 y**, 70% at 7 y. Contralateral GCNIS 5–9%, ≈36% with atrophy/UDT/age <40.
- Microlithiasis alone needs no work-up (matters only with previous GCT).

KEY POINT
GCT is a disease of the arrested gonocyte: UDT, family history and atrophy mark an abnormal developmental field, so orchidopexy reduces but never abolishes risk.

2 · Pathology and classification

- P3 Classify testicular tumors and describe management of seminoma — *WHO 2016 groups, i(12p), any NSGCT element = NSGCT, seminoma IHC*

Subtype	Markers	IHC / histology	Hallmark
Seminoma	hCG ≈15%; **never AFP**	CD117+, CD30−, PLAP+	4th–5th decade; can transform to NSGCT
Spermatocytic	None	OCT3/4−, PLAP−; three cell types	6th decade; never mixed, not with UDT; **benign** unless sarcomatous/anaplastic
Embryonal carcinoma (EC)	AFP/hCG or normal	AE1/AE3+, OCT3/4+, CD30+†, c-KIT−	Totipotent; spreads with normal markers; predicts occult spread
Yolk sac	AFP, not hCG	Schiller–Duval bodies (≈50%), hyaline globules	Pure form: children, mediastinum
Choriocarcinoma	Very high hCG	Syncytio- + cytotrophoblasts	Haematogenous (lung, liver, brain); bleeds; poor risk
Teratoma	Normal/mild AFP	≥2 germ layers; maturity irrelevant postpubertally	Benign histology, malignant genetics; **chemoresistant**

- Teratoma in ≈half of mixed GCTs; unresected → growing teratoma, unresectability, somatic transformation (rhabdomyosarcoma, adenocarcinoma, PNET; late, metastatic sites) or late relapse → surgery integral to cure.

- hCG α-subunit shared with TSH/LH → hyperthyroidism, gynaecomastia. Expert review changes subtype ≈1/3, LVI ≈1/5.

> **KEY POINT**
> Management follows the most chemoresistant element: EC drives occult metastasis, choriocarcinoma haemorrhage, teratoma surgery.

3 · Presentation

- **Painless mass** commonest. Metastases at diagnosis ≈2/3 NSGCT vs 15% seminoma; symptoms in 10–20% (back pain, leg swelling, dyspnoea, left supraclavicular node). Gynaecomastia ≈2%.
- Epididymo-orchitis: re-examine 2–4 weeks after antibiotics. ≈1/3 first misdiagnosed. Man 15–50 with retroperitoneal or left supraclavicular mass → examine testis, US, markers.

4 · Diagnosis and initial management

Scrotal ultrasound (US)

- GCT **hypoechoic**: homogeneous → seminoma; heterogeneous/cystic/calcified → NSGCT. MRI if equivocal.
- Metastatic GCT, normal palpation → US for **burned-out primary** (scar, calcification) → orchidectomy. Normal testes → extragonadal GCT.
- Impalpable <10 mm lesion, normal markers: 20–50% GCT → orchidectomy, inguinal exploration + frozen section, or serial US.

Serum tumour markers

- P1 Tumour markers in urology — *sources, never-raised-in, half-lives, S stage, raised AFP = NSGCT*
- Half-life AFP **5–7 days**, hCG **24–36 h**, LDH 24 h.
- False positives: AFP — liver disease, HCC/gastric cancer, ataxia telangiectasia; hCG — LH cross-reactivity in hypogonadism (normalises 48–72 h after testosterone), marijuana; LDH — muscle injury, lymphoma.
- **Post-orchidectomy values** (within 4 weeks of treatment) set S stage; markers never decide orchidectomy. Slow decline, negative imaging → monitor; **rising → chemotherapy**.
- Unstable, very high AFP/hCG: chemotherapy first, orchidectomy later (testis a sanctuary).
- miR-371a-3p: GCT-specific, not raised by teratoma.

Radical inguinal orchidectomy

- Testis + cord to internal ring within 1–2 weeks. **Trans-scrotal approach/biopsy contraindicated** (inguinal/pelvic spread, local recurrence). Curative alone in 80–85% CS I seminoma, 70–80% CS I NSGCT.
- **Testis-sparing**: synchronous bilateral or solitary testis (<2–3 cm, adequate androgen) or suspected benign; frozen section, biopsies; GCNIS → ≥20 Gy (may defer for paternity).
- Contralateral biopsy if atrophy, UDT, age <40, suspicious US. Extragonadal: raised markers + normal testes diagnostic; else biopsy, i(12p).

5 · Staging

Spread and imaging

- P1 Lymphatic drainage of testis and scrotum, RPLND templates and penile cancer implications — *retroperitoneum first site 70–80%, landing zones, crossover*
- CT abdomen–pelvis for all; false negative in **25–35% of CS I NSGCT** → landing-zone nodes 5–9 mm suspicious; equivocal + normal markers → repeat CT 6–8 weeks.
- Chest X-ray all; chest CT if markers raised, abdominal disease, abnormal X-ray, or CS I NSGCT with LVI/EC. Isolated mediastinal nodes → biopsy.
- Brain imaging if hCG >10,000 IU/L. No routine bone scan; no FDG-PET at diagnosis.
- **Sperm cryopreservation** before RPLND/chemotherapy/radiotherapy (oligospermia in up to half).

TNM staging — the only TNM with a serum (S) category

Category	Definition
pTis	GCNIS
pT1	Testis/epididymis, **no LVI**; may invade tunica albuginea
pT2	LVI, or tunica vaginalis
pT3 / pT4	Spermatic cord / scrotum
N1 / N2 / N3	≤2 cm / >2–5 cm / >5 cm (pN1 ≤5 nodes, pN2 >5 or extranodal; pN invalid after chemotherapy)
M1a / M1b	Non-regional nodes or lung / other viscera
S1	LDH <1.5×N *and* hCG <5000 mIU/mL *and* AFP <1000 ng/mL
S2	LDH 1.5–10×N *or* hCG 5000–50,000 *or* AFP 1000–10,000
S3	LDH >10×N *or* hCG >50,000 *or* AFP >10,000

Stage	Criteria
IA / IB	pT1 / pT2–4, N0 M0 S0
IS	Any pT, N0 M0, **S1–3**
IIA / IIB / IIC	N1 / N2 / N3, M0, S0–1
IIIA	M1a, S0–1
IIIB	N1–3 or M1a with S2
IIIC	N1–3 or M1a with S3, or **any M1b**

IGCCCG risk — metastatic disease

Risk	NSGCT	Seminoma
Good	Testis/retroperitoneal primary, no NPVM, AFP <1000, hCG <5000, LDH <1.5×N (56%; 5-y survival 92%)	Any primary, no NPVM, any hCG/LDH, normal AFP (90%; 86%)
Intermediate	Any of AFP 1000–10,000, hCG 5000–50,000, LDH 1.5–10×N (28%; 80%)	**NPVM** (10%; 72%)
Poor	**Mediastinal primary**, NPVM, AFP >10,000, hCG >50,000 or LDH >10×N (16%; 48%, now ≈71%)	**None**

KEY POINT

IGCCCG uses markers at start of chemotherapy, not pre-orchidectomy, and is invalid at relapse. Seminoma risk depends on NPVM alone; mediastinal NSGCT is always poor risk.

6 · Treatment principles

- Centralise; full-dose cisplatin despite cytopenias/renal impairment; resect all residual disease.

	Seminoma	NSGCT
CS I / II / III	85 / 10 / 5%	≈1/3 each
Occult metastasis CS I	10–15%	25–35%
Radiosensitive	Exquisitely	No (except brain)
Metastatic teratoma	Rare	Frequent → surgery
Markers	Rarely useful	Central

- Seminoma relapses contain NSGCT in 10–15% → biopsy non-responders. Expert nerve-sparing RPLND has the best long-term toxicity profile.

GCNIS management

- Orchidectomy, radiotherapy **≥20 Gy** (<20 Gy → recurrence; spares Leydig cells, yet up to 40% need testosterone), or observation.
- Subfertile → serial US or sperm retrieval first. Normal contralateral testis wanting paternity → orchidectomy (scatter). Cisplatin partly clears GCNIS → biopsy ≥2 y later or radiotherapy.

7 · NSGCT management

CS I NSGCT

- Occult metastasis 20–30% → adjuvant overtreats 70–80%.
- **Risk factors**: **LVI** (strongest), **EC predominance**; high risk ≈50% relapse, low <20%.
- **Surveillance**: relapse ≈28%, >90% within 2 y, 65–75% retroperitoneal; chemo-naive, curable.
- **Primary RPLND**: removes teratoma (15–25% of occult nodes); pathological stage II 19–28%, mostly cured by RPLND alone; retroperitoneal recurrence <2% (no routine abdominal CT). pN1 → observe; pN2–3 (relapse >50%) → **2 cycles adjuvant** (relapse ≤1%) or observe.
- **BEP×1**: lowest relapse (≈2–3%) but leaves teratoma, exposes all to late toxicity; relapses chemoresistant.
- Template: full bilateral nerve-sparing (ejaculation >90%) standard, mandatory for suspicious nodes; modified acceptable for clinically negative nodes.

ALGORITHM

CS I NSGCT → low risk (no LVI, no EC predominance) → surveillance → high risk → surveillance or BEP×1 or nerve-sparing RPLND (expert), by preference and toxicity

KEY POINT

Surveillance maximises quality-adjusted survival when relapse risk <≈1/3. All options cure ≈100%; they differ in overtreatment vs difficulty of salvage.

CS IS, IIA–B

- **CS IS** → induction chemotherapy.

- **CS IIA–B**: raised markers or nodes **>3 cm** → chemotherapy (± PC-RPLND). Normal markers, nodes <3 cm → **primary RPLND**: 13–35% pN0, ≈30% teratoma, half avoid chemotherapy.

CS IIC–III: induction by IGCCCG

- P2 CRE & BEP — *BEP×3/EP×4 good, BEP×4 (VIP if lungs poor) intermediate/poor, toxicities*
- BEP×3 = BEP×4; EP×3 and carboplatin inferior. Second line TIP×4 / VeIP×4 / VIP (TIP best for late relapse). HDCT (carboplatin–etoposide + stem cells): three negative RCTs (two first-line) → specialised centres; carboplatin replaces cisplatin because nephro-/neurotoxicity limits escalation.
- Ifosfamide: haematological, urological toxicity. Slow marker decline adverse; intensification improved PFS not OS → BEP×4 stays standard.

Post-chemotherapy residual masses (NSGCT)

- Normal markers + residual mass **>1 cm** → **PC-RPLND**, full bilateral template. Necrosis ≈40%, teratoma ≈45%, viable ≈15%.
- No model, imaging or **FDG-PET** (teratoma not avid; ≈20% false negative) predicts necrosis → always resect.
- Complete response (≤1 cm): observe, or resect if pre-chemotherapy >3 cm — controversial.
- Multiple sites: resect all, retroperitoneum first (necrosis there predicts necrosis elsewhere). Somatic transformation 6–8% → complete resection.
- Adverse after viable tumour: intermediate/poor risk, incomplete resection, >10% viable cells. Inadequate PCS caused most deaths in the French BEP×3 vs EP×4 trial.
- Marker-positive partial response or progression → second-line chemotherapy.

Relapsed NSGCT

- P1 Growing teratoma syndrome — *enlarging mass, normal markers → stop chemotherapy, resect*
- **Chemo-naive** → induction by IGCCCG, cure >95%; small marker-negative retroperitoneal relapse → RPLND option.
- **Early (≤2 y) post-chemotherapy** → TIP/VeIP/VIP×4 or HDCT; residual masses → post-salvage surgery.
- **Desperation surgery**: rising markers, single resectable site; survival 33–57%.
- **Late relapse (>2 y)** ≈3%; yolk sac commonest, chemoresistant → **complete resection**; TIP if needed; biopsy if markers normal.

8 · Seminoma management

- P3 P2 Classify testicular tumors and describe management of seminoma; Describe anatomy of lymphatic drainage of testis and scrotum. Describe management of stage one seminoma of testis. — *CS I options table, risk factors, Calvert, stage II–III, residual mass, follow-up*
- CS I ≈80%. Risk factors tumour **>4 cm**, rete testis invasion (12/16/32%) — not validated, so surveillance preferred for all.
- Radiotherapy: dog-leg if prior scrotal/inguinal surgery; scatter oligospermia 8%, SMN ≈18% at 25 y; pelvic nodes only 1–3% without prior surgery. Carboplatin ×2 relapse ≈1.5%; relapse after carboplatin treated as chemo-naive.
- Surveillance harder than NSGCT: hCG detects <5%, examination/X-ray almost none; median relapse 14 months (NSGCT 4–8).
- IIA–B nonbulky (≤3 cm): DL 25–30 Gy + 5–10 Gy boost, BEP×3/EP×4, or primary RPLND (SEMS, COTRIMS; ≈10% relapse). Bulky/multiple → chemotherapy.

- Residual mass: necrosis ≈90%; complete resection 58–74%; many regress over 12–18 months. Viable: >3 cm 13–55%, <3 cm 0–4%.

> **ALGORITHM**
> Residual seminoma mass → <3 cm → observe → >3 cm → FDG-PET ≥6 weeks after chemotherapy (earlier = false positives) → negative → observe → positive → template PC-RPLND (or repeat PET/observe); mass-only excision → 33% local relapse

- Relapse after radiotherapy → first-line chemotherapy; after chemotherapy → VeIP×4 or HDCT; normal markers → biopsy (transformed NSGCT). Late seminoma relapse may be favourable.

9 · Brain metastases

- ≈1% at diagnosis; **choriocarcinoma/very high hCG**; fatal haemorrhage risk. BEP×4 → resect residuals; radiosurgery if unresectable. Brain relapse after complete response is worse.

10 · Treatment sequelae and follow-up

- RPLND: ejaculatory loss (>80% full template without nerve sparing), small-bowel obstruction. Orchidectomy alone: hypogonadism 10–20%.
- Chemotherapy: toxic death 0–2.4% (good), 3–4.4% (intermediate/poor); azoospermia near universal, ≈half recover by 2 y, 80% by 5 y.
- Late: Raynaud 20–45%, neuropathy 14–43%, ototoxicity, hypogonadism 15–40%; **cardiovascular disease** and **SMN** (60–100% excess after cisplatin or radiotherapy, 200% after both).
- Follow-up most intensive years 1–2 (NSGCT) / 1–3 (seminoma); seminoma surveillance beyond 5 y; after carboplatin ≥5 y; after full RPLND chest imaging + markers. Survivorship†: BP, lipids, glucose, testosterone.

> **KEY POINT**
> Because cure is expected, late cardiovascular disease and second cancers — rising with each added modality — decide between equally curative options.

11 · Non-germ-cell tumours†

- †**Leydig cell**: commonest sex cord–stromal (≈1–3%); children → precocious puberty; adults → gynaecomastia. Reinke crystals; inhibin, calretinin, Melan-A +. ≈10% malignant (>5 cm, necrosis, LVI, mitoses) → chemo-/radioresistant → RPLND for nodes. Small marker-negative endocrine lesion → testis-sparing excision.
- †**Sertoli**: mostly benign; large-cell calcifying variant with Carney complex, Peutz–Jeghers (bilateral, gynaecomastia).
- †**Lymphoma**: commonest in men **>50–60 y** and commonest bilateral; DLBCL → orchidectomy, R-CHOP + CNS prophylaxis + contralateral testis radiotherapy.

12 · Landmark trials

Trial	Design	Result	Take-home
IGCCCG 1997	Pooled 5202 NSGCT, 660 seminoma	Markers, NPVM, mediastinal primary predict outcome	Risk groups select regimen
PVB vs BEP	RCT, advanced	Equal survival, less neurotoxicity	BEP standard

Trial	Design	Result	Take-home
BEP×3 vs ×4	RCT, good risk	Equal survival	BEP×3
VIP vs BEP×4	RCT, intermediate/poor	Equal, more toxicity	VIP only if lungs at risk
MRC TE19/EORTC 30982	Carboplatin ×1 vs radiotherapy, CS I seminoma	5-y RFS 94.7% vs 96%	Equivalent
MRC 20 vs 30 Gy	RCT, CS I seminoma	Equal relapse, less toxicity	20 Gy PA
Albers (German)	BEP×1 vs unilateral RPLND, CS I NSGCT	2-y PFS 99% vs 92%	Favours BEP×1
Williams	Adjuvant ×2 vs observation, pathological stage II	Relapse 6% vs 49%; same survival	Both acceptable
MRC surveillance	2 vs 5 CTs, CS I NSGCT	No difference	Fewer scans safe

80 Surgery of Testicular Tumours

1 · Evaluation of the testis mass

- Urgent: delay escalates the intensity of curative therapy.
- History → examination (supraclavicular, breasts, abdomen, scrotum) → scrotal US (key role: **contralateral testis**; bilateral synchronous ≈ **1%**) → markers (AFP, β-hCG, LDH: diagnosis, stage, post-orchidectomy baseline).
- Discuss prosthesis; † offer sperm banking before orchidectomy.

2 · Radical inguinal orchidectomy

Treatment of choice: **early cord control** before handling the testis; removes testis, epididymis, cord to internal ring. † Scrotal approach breaches a second lymphatic field (inguinal) → local recurrence.

> **ALGORITHM**
> 3–5 cm incision over canal → open external oblique → **preserve ilioinguinal nerve** → mobilise cord at pubic tubercle, **Penrose early clamp** → deliver testis, divide gubernaculum → dissect cord to internal ring.

- **High ligation**: vas and gonadal vessels ligated **separately**, **non-absorbable** suture, **1–2 cm** tail — marks stump for excision at RPLND (vas need not be removed).
- Scrotal support 2–3 days. † Prosthesis can go in at the same sitting.

> **KEY POINT**
> † **Scrotal violation**: inguinal exploration, high cord ligation, excise scrotal scar; routine hemiscrotectomy no longer needed; chemotherapy-era outcomes not compromised.

Partial orchidectomy (testis-sparing surgery)

- P4 Testis sparing surgeries for CA testis — *indications (solitary/bilateral, small, polar), intraoperative US, cold ischaemia <30 min, frozen section, GCNIS 18–20 Gy, recurrence → completion orchidectomy*
- **All three**: polar, **≤2 cm**, contralateral testis compromised/absent. **Never elective with a normal contralateral testis.**
- Masses **<5 mm**, negative markers → low malignancy; partial orchidectomy or selective surveillance.
- Cold ischaemia omissible if **<30 min**. Tunica incision vertical for ventral midline, horizontal for medial/lateral lesions (spares segmental arteries); three-point fixation.
- GCNIS: completion orchidectomy vs **18–20 Gy** vs observation (local recurrence ≈ 7.5%). **Local recurrence → completion orchidectomy**, always.

Delayed orchidectomy

Life-threatening metastatic GCT → chemotherapy first, then **delayed orchidectomy in all** (testis sanctuary; **discordant response**); usually teratoma or necrosis. Apparent extragonadal retroperitoneal primary: intratubular neoplasia in up to 42%; lateralising nodes → orchidectomy that side. Staging: CT (MRI alternative), no FDG-PET; repeat markers.

3 · RPLND — terminology and indications

Term	Indication
Primary RPLND	High-risk CS I or low-volume stage II NSGCT, **normal markers**
PC-RPLND	After induction; residual **>1 cm**, normal markers
Salvage PC-RPLND	After second-line chemotherapy
Desperation	Markers still elevated; resectable
Redo	Retroperitoneal recurrence after prior RPLND
Late relapse	**>24 months** after complete response

Each step down = more chemoresistant disease, harder surgery.

Preoperative planning

- CT within **6 weeks**, markers within **7–10 days**; vascular variants; no bowel prep.
- **Bleomycin**: PFTs → alert anaesthesia → **low FiO_2, restrictive fluids**.
- IVC thrombus: total occlusion → IVC resection; partial → reconstruction.

4 · Templates

- P1 Lymphatic drainage of testis and scrotum, RPLND templates and penile cancer implications — *landing zones, bilateral infrahilar and modified templates, omit suprahilar/interiliac, 3–23% out-of-template, PC-RPLND full bilateral; modified only if ≤5 cm in landing zone + normal markers + good/intermediate risk; L1–L4 fibres to hypogastric plexus*
- Bilateral template limits: renal vessels, ureters, common iliac arteries. No randomised trial of template extent. Modified PC template lesion **≤5 cm**.

5 · Open RPLND technique

- **Transperitoneal** or **extraperitoneal** (bowel contained → less ileus).
- Incise caecum to IMV, line of Toldt to foramen of Winslow; free duodenum/pancreas; **liberal clips** against chyle leak.
- **Split and roll**: split at 12 o'clock on aorta below left renal vein → caudally to IMA; ligate cephalad lymphatics; roll packets off; ligate lumbars.
- Lumbar arteries usually 3 pairs; **2nd pair at IMA level**. **Common lumbar vein** largest. **Right sympathetic trunk** posterolateral to IVC mistaken for lymphatic tissue; spare genitofemoral nerve.
- Gonadal vein excised to internal ring and orchidectomy suture.

Nerve sparing

- Left fibres: lateral aortic border, under left renal vein, around left common iliac — near gonadal artery and IMA. Right fibres: emerge **beneath IVC**, oblique over aorta — at risk ligating superior/common lumbar veins.

> **KEY POINT**
> Identify fibres **prospectively, before dividing anything**; loop them, pass packet beneath. Avoid traction (neuropraxia → transient anejaculation).

Postoperative care

- Optionally close mesenteric root (volvulus). No routine drain (if used: remove at **<100 mL/24 h** on diet); no NG tube; low-fat diet days 1–2; mobilise day 1; home day 3–5.

6 · Auxiliary procedures (PC-RPLND 23–45%)

- **Nephrectomy** (commonest; **left**): strongest predictor residual **>10 cm**; risk rises fibrosis < teratoma < viable, and in salvage/desperation/redo. Concern: future cisplatin eligibility.
- **Vascular**: plan in every PC-RPLND (subadventitial aortic injury → fatal bleed). Patch or PTFE/Dacron graft.
- **IVC**: narrowing **<25%** → venorrhaphy, >25% → patch/graft; ligation acceptable (collaterals); grafts thrombose; mass >5 cm → ≈20% need IVC work. **Aorta never ligated**; duodenal injury + graft → aortoduodenal fistula.
- **Liver**: discordant ≈half; resect, or microwave ablation if **<3 cm** unresectable. **Thorax**: 25–50% discordant → RPLND first; resect >1 cm unless retroperitoneum fibrosis only (predicts thoracic necrosis ≈89%). **Pelvis**: mostly teratoma. **Retrocrural**: transdiaphragmatic along right crus.

7 · Decision making

- **Complete remission (≈70%)**: generally observe (≤1 cm harbours teratoma ≈ 1/5, viable ≈ 1/20).
- **Necrosis predictors**: no teratoma in primary, normal pre-chemotherapy AFP/β-hCG, raised LDH, >50% shrinkage, residual <2 cm.
- **After primary RPLND**: pN0/teratoma → observe; pN1 → observe; pN2 → 2 cycles vs observe. Observation cures ≈70%; relapses need ≥1 extra BEP or 2 extra EP.

8 · Histology of residual mass after chemotherapy

Finding	Frequency	Outcome/action
Fibrosis/necrosis	**≈40%**	RFS/CSS ≈95%; no adjuvant
Teratoma	**≈45%**	RFS 80–90%; must be resected
Viable GCT	**≈15%**	Survival 50–70%; consider 2 adjuvant cycles

> **KEY POINT**
> **Resect every NSGCT residual >1 cm**: no imaging (including PET †) or marker separates necrosis from teratoma; teratoma is chemoresistant, grows, relapses late, transforms. Surgery is staging and the only treatment; histology decides further therapy.

- **Fizazi** (viable GCT): incomplete resection, ≥10% viable cells, IGCCCG intermediate/poor → 0/1/≥2 = 5-y OS 100/83/51%.
- Adjuvant: NCCN — incomplete resection or intermediate/poor; EAU — incomplete resection. Not after salvage chemotherapy.

9 · High-risk PC-RPLND

- **Salvage**: viable GCT ≥doubled; OS 60–75%. After HDCT viable ≈40%, still worthwhile.
- **Desperation**: OS ≈54%. Operate if confined, completely resectable, markers declining/plateaued, or chemotherapy exhausted; **not** with rapidly rising markers or multisite

progression. Plateau → >75% fibrosis/teratoma. **Post-op marker normalisation** is the only OS predictor.

- **Redo**: technical failure (retroaortic/retrocaval tissue, lumbars) → **complete great-vessel mobilisation first time**.
- **Late relapse**: yolk sac, somatic-type malignancy → **surgery first**.

10 · Unique situations

- **Seminoma residual**: desmoplasia, no teratoma, viable ≈10%. **<3 cm** → observe; **>3 cm** → FDG-PET (≥6 weeks †); avid → PC-RPLND (select), salvage chemotherapy or HDCT.
- **Sex cord stromal**: malignant features >4 cm, necrosis, mitoses, atypia, infiltration, LVI; RPLND of unproven benefit. Leydig almost always stage I; Sertoli more aggressive.

11 · Outcomes, fertility and complications

- RPLND spares spermatogenesis; 40–60% abnormal semen at diagnosis.
- **Ejaculation**: template or nerve sparing → antegrade **>90%** (primary), no oncological cost. † Loss → pseudoephedrine/imipramine, post-ejaculatory urine sperm, electroejaculation, TESE.
- Complications: primary 10.6–24%, PC up to 56%. Mortality primary ≈0, PC <1%.

Complication	Key facts / management
Pulmonary (bleomycin)	PC 3–5%, ARDS. Low FiO_2, restrictive fluids; tachycardia alone no fluid trigger; accept SpO_2 **88–90%**
Ileus	Up to 26% (commonest); extraperitoneal, no NG tube, early feeding
Chylous ascites	Primary 0.2–2.1%, PC 2–12%; suprahilar dissection (cisterna chyli L1–L2)
Lymphocele	Symptomatic ≤1.7%; air/rim enhancement = infected → drain ± antibiotics
VTE	Primary <1%, PC 0.1–3.1%; pharmacological if prior chemotherapy/VTE/obesity
Neurological	Femoral neuropraxia; paraplegia with bulky disease

- P1 Lymphocele and postoperative lymphatic leak (chylous ascites) — *milky tap, triglyceride 2–8× serum, protein >3 g/dL; stepwise management*

> **ALGORITHM**
> Paracentesis → drain if reaccumulates → very-low-fat, high-protein diet + MCT + octreotide → NBM + TPN → refractory: lymphangiography ± embolisation, peritoneovenous shunt or exploration → low-fat diet ≥ **4 weeks** after resolution.

12 · Landmark trials

Trial	Design	Result	Take-home
Williams (Testicular Intergroup)	RCT, pathological stage II: 2 adjuvant cycles vs observation	Adjuvant near-abolished relapse; equal survival	Observe pN1; individualise pN2
SEMPET	FDG-PET in post-chemotherapy seminoma residuals	>3 cm: specificity 100%, NPV 96%; ≤3 cm PPV 37%	PET for >3 cm only

81 Minimally Invasive Retroperitoneal Lymph Node Dissection

- P3 P4 Minimal invasive retroperitoneal lymph node dissection; Robotic RPLND — *indications, templates, nerve sparing, robotic steps, lap vs robot, outcomes, AUA/EAU positions, miR-371a-3p*

1 · Indications and rationale

- Goals in early-stage disease: preserve cure, avoid **dual therapy**, minimise lifelong toxicity.
- **Primary**: marker-negative CS I NSGCT; small-volume stage II NSGCT and seminoma. Stages and cures occult disease; spares chemotherapy in the **20–30%** with enlarged nodes who prove pN0.
- **PC-RPLND**: NSGCT residual **>1 cm**; select seminomas.
- MI-RPLND = latest de-escalation step after dropping suprahilar dissection, nerve-sparing, modified templates. Robot: fourth-arm retraction, **circumferential great-vessel dissection with lumbar control**.

KEY POINT
Much recovery gain came from ERAS pathways, which help every approach — the benefit of access alone is smaller than it looks.

2 · Guidelines and controversy

- NCCN silent; AUA (early-stage) **may be offered** by GCT/MIS-expert surgeon, patient told long-term data lacking; EAU **specialist centres** expert in open and MIS only.
- No noninferiority evidence → most controversial operation in urological oncology.
- Concerns: staging-only dissection (no lumbar division, low yields, excess adjuvant chemotherapy even pIIA → dual therapy; residual teratoma → late relapse); wrong patients (stage IA operated; pN0 up to 83%); console great-vessel surgery, surgeons lacking open skills, conversion **≈5%**; no series > 5 years, peritoneal carcinomatosis reported.

3 · Patient selection

- No absolute contraindications; relative: extensive prior surgery, morbid obesity.
- **Suits MIS**: chemo-naive high-risk IB; small-volume II (1–2 landing-zone nodes, none **>3 cm**); PC solitary landing-zone mass with small response (suggests teratoma), little desmoplasia.
- **Favour open**: PC-RPLND in general, large mass, large pre→post chemotherapy shrinkage (desmoplasia), possible vessel reconstruction, somatic malignancy, retrocrural disease.

4 · Unchanging operative principles

- Imaging within **≈4 weeks**, markers within **1–2 weeks**.
- **MI-RPLND must completely duplicate open RPLND** — template, completeness and nerve-sparing unchanged.

ALGORITHM
Safe entry → exposure → **split-and-roll** → prospective nerve sparing → **lumbar vessel division** → spermatic cord removal → closure

- Chemo-naive: extended unilateral nerve-sparing (debated); enlarged nodes at surgery → bilateral. Post-chemotherapy: **bilateral full template** standard.

- Incomplete RPLND/mass excision worsens survival.

5 · Preparation, positioning and ports

- Haemorrhage plan; bleomycin → careful fluids/oxygen; **open set in theatre**. High-fat meal day before highlights lymphatics; no bowel prep. Pad pressure points (rhabdomyolysis).
- Lateral decubitus (≈45°): familiar, bilateral needs repositioning. Laparoscopic supine: four 12-mm midline trocars. Robotic steep Trendelenburg: camera 4–6 cm below umbilicus, any template; **peritoneal hammock** (posterior peritoneum incised from Treitz round caecum, edges sutured to anterior wall).

6 · Operative steps

- **Right**: Toldt line to foramen of Winslow, **Kocherise**; gonadal vein divided at IVC; split over IVC from iliac crossing to left renal vein; clip retropancreatic lymphatics; roll paracaval packet, spare fibres, divide lumbars (Weck clips vessel side). **Do not ligate the right renal artery** in the interaortocaval space; preserve accessory lower-pole arteries.
- **Left**: gonadal vein at left renal vein; control **lumbar vein into left renal vein**; IMA may be ligated for mesocolic rotation.
- Lower pneumoperitoneum to check haemostasis; bag packets; **no routine drain**; discharge **1–2 days**.

7 · Complications and management

> **ALGORITHM**
> Major vascular injury → pressure + raise pneumoperitoneum → inform anaesthesia, blood, vascular help → uncontrolled: keep pressure, undock, convert

- MIS repair: **identify injury before acting — no blind clipping**.
- Chylous ascites: liberal clips at left renal vein and IMV crossing (energy devices unreliable); rates match open. † Treat: low-fat/MCT diet → paracentesis, octreotide, TPN → ligation.
- Antegrade ejaculation **≥90%** after primary nerve-sparing.

8 · Outcomes vs open

- **No randomised trial**. Shorter stay, less blood loss, longer operative time; brisk bleeding the unique risk.
- Robotic node yields approach open benchmark **≈30**; adjuvant chemotherapy still too frequent.
- Benchmark recurrence-free ≈**85%** for limited pN+ without adjuvant chemotherapy. Low MI recurrence **confounded** by high pN0 and adjuvant rates. PC data too small.

9 · Landmark trials

Trial	Design	Result	Take-home
GETUG T93BP	Good-risk NSGCT: BEP×3 vs EP×4	Most relapses/deaths without or with incomplete RPLND	Complete RPLND matters

82 Tumours of the Penis

1 · Overview

- SCC = **~95%** of penile malignancy. **0.4–0.6%** of male cancers in North America/Europe; up to **10%** in parts of Asia, Africa, South America.
- Mean age mid-50s (rise in 6th decade); a meaningful minority <40. Prognosis rests on **primary stage/histology** and **nodal metastasis**.

2 · Premalignant lesions — PeIN

- P1 Premalignant conditions in urology (including premalignant lesions of penis) — *PeIN definition, WHO 2022 all high grade, HPV-associated vs differentiated, E6/E7/p16, EQ/Bowen/Bowenoid table, LS 2.3–9%, Buschke–Löwenstein, organ-preserving treatment*
- EQ: glans, uncircumcised, red velvety; mimic balanitis; progression **10–33%**. Bowen: shaft scaly plaque (mimics psoriasis, eczema, superficial BCC); **~5%**. Bowenoid papulosis: shaft papules, young men, HPV; **does not progress**.
- Eponyms describe site/look only. EQ vs balanitis: full-thickness atypia, bulbous rete, capillary proliferation, plasma cells. Metastasis from PeIN extremely rare.
- Differentiated PeIN hard to separate from pseudoepitheliomatous hyperplasia. Imiquimod = topical choice for condyloma.

3 · Aetiology and risk factors of SCC

Factor	Key point
No neonatal circumcision	Neonatal circumcision virtually eliminates invasive cancer; protects against invasive SCC, **not CIS**. Adult circumcision little/no protection
Phimosis	Present in **25–75%**; **7–10×** risk; mediates the non-circumcision risk (smegma → inflammation → fibrosis)
HPV	Low risk 6, 11; high risk **16, 18**, 31, 33, 45, 52, 58; **HPV 16 commonest**. Present in **~50%**; wives ~3× cervical cancer risk
HIV	~4× risk of penile cancer death; PeIN progression ~6 years earlier
Tobacco	All forms; dose-dependent, independent of phimosis
Lichen sclerosus	Cancer **2.3–9%**; HPV-**independent** SCC
Trauma, obesity	Scarred shaft; buried penis/functional phimosis
† **PUVA**	† Genital exposure markedly increases SCC → shield genitalia

- HPV-independent pathway: CDKN2A hypermethylation, TP53 mutation, MDM2; inflammation → ROS, COX-2.
- **Not linked**: occupation, gonorrhoea, syphilis, herpes, alcohol, marijuana.
- **Prevention**: neonatal circumcision (AAP 2012: benefits outweigh risks), hygiene, no tobacco, HPV vaccination (nonavalent adds 31/33/45/52/58).

> **KEY POINT**
> Two routes converge on the preputial sac: HPV-driven (E6/E7, p16+, basaloid/warty) and inflammation-driven HPV-independent (phimosis, smegma, LS). Neonatal circumcision, hygiene and vaccination each prevent part; adult circumcision almost none.

4 · Natural history and spread

- Glans (~½), prepuce, coronal sulcus; shaft <2%. **Flat ulcerative tumours metastasise earlier**.
- **Buck fascia** a temporary barrier; through it and tunica → corpora → vascular spread.
- Distant metastases uncommon (**1–10%**), late. Untreated: death within **2 years** (femoral erosion haemorrhage, sepsis); no spontaneous remission.
- P1 Lymphatic drainage of testis and scrotum, RPLND templates and penile cancer implications — *penile superficial/deep lymphatics, stepwise bilateral spread, no skips*

5 · Presentation and diagnosis

- Induration, warty growth, ulcer with rolled edges; may hide under phimosis. Pain slight. 15–50% delay >1 year.
- Examine size, fixation, corpora, base, scrotum, perineum; **bilateral inguinal palpation**.
- **Biopsy mandatory before definitive therapy** — depth, grade, predicts occult nodes; superficial biopsies mislead. Dorsal slit, or frozen section then immediate resection.
- Labs: anaemia, leucocytosis, hypoalbuminaemia. **Hypercalcaemia without bone metastases** (PTH-related; bulky inguinal disease; may resolve after node excision): saline → bisphosphonate → calcitonin; diuretic only if overloaded.
- Differential: condyloma, BXO, chancre, chancroid, herpes, LGV, granuloma inguinale, TB.

6 · Histology

Subtype (WHO 2022)	HPV	Features	Nodal risk
Usual (keratinising) **45–65%**	Independent	Usually grade II; LVI/PNI ~⅓	25–40%
Warty	Associated	Cauliflower, fibrovascular cores; slow	17–25%
Basaloid	Associated	Small cells, necrosis, deep	**>½ node positive**
Papillary	Independent	Low grade, no koilocytosis	≤12%
Verrucous	Independent	**Broad pushing border**	**No metastasis**; recurs if under-excised
Sarcomatoid	Independent	Spindle cells	**Most aggressive**
Rare: pseudoglandular, adenosquamous; clear cell, lymphoepithelioma-like (HPV)	—	Deep invasion	Pseudoglandular, clear cell worst

- **Any anaplastic cells = grade III** (half heterogeneous). **LVI** predicts nodes in every study; **PNI** strong.
- **hrHPV/p16+ = better DSS** (not nodal status). Adverse: p53, PD-L1, CRP, NLR; nomograms unvalidated.

7 · Imaging

- **Primary**: examination reliable; US sees tunica but cannot separate T1/T2; **MRI** ± artificial erection (PGE1) if equivocal and organ sparing planned; CT not useful.

- **cN0 groin**: **no imaging detects micrometastases** → surgical staging; CT if obese/prior groin surgery.
- **cN+ / T3–T4**: CT chest/abdomen/pelvis (central necrosis, irregular border → poor pathology); **FDG-PET/CT** best for pelvic/distant.

8 · Staging — TNM 8th edition (2017)

T	Definition
Tis	PeIN
Ta	Non-invasive localised SCC (incl. pure verrucous without destructive invasion)
T1	Glans: lamina propria · Foreskin: dermis/lamina propria/dartos · Shaft: tissue between epidermis and corpora
T1a	**No LVI, no PNI, not high grade**
T1b	**LVI and/or PNI, or high grade (grade III/sarcomatoid)**
T2	**Corpus spongiosum** ± urethra
T3	**Corpora cavernosa** (incl. tunica albuginea) ± urethra
T4	Scrotum, prostate, pubic bone

N	Clinical (cN)	Pathological (pN)
1	Palpable mobile unilateral node	**≤2 unilateral**, no ENE
2	Mobile ≥2 unilateral or bilateral	**≥3** unilateral or bilateral
3	Fixed mass or pelvic nodes	**ENE or pelvic nodes**

Stage	I	IIA	IIB	IIIA / IIIB	IV
TNM	T1a N0	T1b, T2 N0	T3 N0	T1–3 N1 / N2	T4, any N3, any M1

- vs 7th: PNI added to T1b; T2/T3 split (cavernosal worse); **urethral invasion no longer upstages**; pN1 widened to 2 nodes.
- M work-up: LFTs, **calcium**, CT, bone scan or FDG-PET/CT as indicated.

> **KEY POINT**
> Traps: T1b and T2 share stage IIA; any N3 or T4 = stage IV without metastases; cN counts palpable nodes, pN counts metastases (pN1 allows two). Primary staged by tissue, groin by surgery.

9 · Management of the primary tumour

- **Penectomy** = oncological gold standard (local recurrence **0–8%**); >half are ≤60 → preserve where safe.
- **Organ preservation**: PeIN, Ta, T1, grade I–II, **compliant**. **Against**: **≥4 cm**, grade III, deep cavernosal invasion, unfit for salvage.
- Higher local recurrence, but **early recurrence salvaged without survival cost** — except pT3/high grade, where relapse worsens survival.
- **Margins**: microscopic spread **~5 mm** (grade I–II), **~10 mm** (grade III); only **<1 mm** raised recurrence → **negative frozen-section margin (>1 mm) enough**; 2-cm dogma abandoned.

Stage / site	Treatment
PeIN, glans	Laser, resurfacing; topical alternative
Ta / PeIN — foreskin, shaft skin	Excision to negative margin; laser; topical (Tis only)
Ta, T1 glans	Excision, resurfacing, glansectomy, RT (not Ta)
T2 glans, no gross cavernosal involvement	**Total glansectomy** ± distal corporal transection; partial penectomy; RT
T2 with corporal invasion, T3	**Partial or total penectomy**
T4	Neoadjuvant chemotherapy → surgery in responders
Recurrence after conservation	Re-excision or penectomy; superficial low-grade re-treated conservatively

- **Glans resurfacing**: strip epithelium/subepithelium off spongiosum (total better than partial) → split-skin graft; erections preserved.
- **Laser**: CO_2 ~0.1 mm (recurrence up to 50%); **Nd:YAG** (1064 nm, up to 6 mm) preferred with base biopsies; recurrence <10% CIS, 10–25% T1.
- **Mohs**: recurrence ~⅓; no clear advantage over frozen-section excision.
- † Partial penectomy: stump for standing micturition; total: crura, **perineal urethrostomy**.

KEY POINT

Margin width is a function of grade, not a fixed 2 cm. Low-grade tumours spread millimetres and recurrences are salvageable, so preservation is safe; in high-grade or cavernosal disease local relapse is lethal and amputation regains its place.

10 · Inguinal nodes *→ Ch 84*

- P2 P3 Importance and management of Inguinal lymph nodes in CA penis; Prognostic index score in carcinoma penis — *nodes = strongest prognostic factor, Kroon early vs delayed, risk groups, DSNB, FNAC without antibiotic trial, PLND >2 nodes/ENE, neoadjuvant TIP, ILND morbidity*
- Cure ~**80%** if ≤2 unilateral nodes, no ENE/pelvic; pelvic positive → ~**14%** 5-y with surgery alone.
- Occult nodes: PeIN/Ta/T1a **0–18%**; T1b+ one-third to >half; corporal ~60%.

ALGORITHM

cN0 low risk → surveillance → node → FNAC → ILND + contralateral staging.

cN0 high risk (T1b+) → bilateral **DSNB** or superficial/fascia-sparing/MI ILND → positive → completion ILND.

cN1–2 mobile <4 cm → biopsy → ILND + contralateral staging; PLND if >2 nodes or ENE.

Fixed / ≥4 cm / pelvic → neoadjuvant TIP (cisplatin-eligible) or chemoradiation → resect responders.

Adjuvant if >2 nodes, bilateral, ENE or pelvic nodes.

- DSNB refinements cut false-negatives ~18% → ~7%. Morbidity: radical > superficial/fascia-sparing > MI > DSNB. No pelvic skip metastases; PLND mainly staging. Reduce morbidity: transverse incisions, saphenous sparing, drain removal by volume, thromboprophylaxis; sartorius transposition *increases* complications.

11 · Radiotherapy

- **Circumcise first** (exposes lesion, prevents oedema/phimosis). Surgical salvage remains curative.
- **EBRT**: **≥66 Gy in 2-Gy fractions over ~6.5 weeks**; wax/Perspex block as **bolus**; ~55% recurrence-free at 5 y. Better: >60 Gy, ≤45 days, ≥2 Gy/fraction; worse: T3, >4 cm, high grade.
- **Brachytherapy** (Ir-192): **T1–T2 glans <4 cm**; ~60 Gy over ~5 days; ~79% relapse-free, preservation ~85% at 5 y; better erections than EBRT. Meatal stenosis (>60 Gy, >25 cc), ulceration (>65 Gy).
- **Late effects**: **meatal stenosis** (10–45%, 18–24 months), ulceration; **1 in 5 local recurrences after 5 years**. Nodal staging indications unchanged after RT (biopsy supplies grade/LVI).
- Surface mould: superficial/non-invasive tumours only. Ulceration: conservative care, hyperbaric oxygen, biopsy if suspicious.
- Groins: **no prophylactic RT** for cN0. Adjuvant RT for **≥2 nodes or ENE**: ≥50 Gy, **54 Gy ENE, 57–60 Gy gross residual**. Chemoradiation (cisplatin if GFR >50 mL/min) for inoperable nodes.

12 · Systemic therapy

- No completed RCTs. Most active: **TIP**; TPF (toxic); doublets for less fit. **Bleomycin regimens discouraged**.
- **Neoadjuvant TIP ×4** for bulky N2–N3 or pelvic nodes: response ~50%; DFS in ~⅔ of responders → **response selects for surgery**.
- Adjuvant chemotherapy: no proven OS gain; consider pN3/pelvic. Chemoradiation (mitomycin + capecitabine) if cisplatin-ineligible.
- Palliative platinum first line (~3× median survival; TIP/TPF unsuitable with neuropathy or low GFR); second line ≤6 months; anti-EGFR sporadic; checkpoint inhibitors ~1 in 6 — trials. InPACT testing sequencing.

13 · Follow-up

- Organ preservation: close follow-up + self-examination (late recurrence after RT).
- Observed groins: 2–3-monthly years 1–2, 4–6-monthly year 3, then 6-monthly, annually. Only ~⅓ of cN0 groin relapses survive 5 y. Second primaries 5–15%.

14 · Non-squamous penile tumours (~5%)

Tumour	Key features / treatment
BCC	Proximal shaft near base; almost never metastasises → excision/Mohs
Melanoma	<1%; glans; haematogenous + lymphatic spread; AJCC cutaneous staging. Circumcision / partial / total penectomy by site; bilateral modified ILND if Breslow ≥1 mm, ulceration, Clark IV–V (DSNB reasonable); adjuvant nivolumab stage IIIB–IV
Sarcoma	Malignant **proximal**; vascular commonest; mimics Peyronie or priapism; **local recurrence characteristic** → superficial → wide excision/partial, deep corporal → total; no ILND unless palpable

Tumour	Key features / treatment
Extramammary Paget	Penoscrotal eczematoid plaque; CK7+, CEA+, S-100−; ~½ have another cancer → screen. Wide excision (up to 3 cm) by **mapping biopsies**
Lymphoma	Usually systemic → R-CHOP even if isolated
Metastases	**Bladder, prostate, rectum**; **priapism** commonest sign; most die within a year (prostate origin longer) → palliative penectomy for pain/solitary distal disease, dorsal nerve section, suprapubic cystostomy

15 · Landmark trials

Trial	Design	Result	Take-home
Neoadjuvant TIP (MD Anderson)	Phase II; TIP ×4, cN2–N3 M0	Response 50%; median OS 17.1 months; durable mainly in responders	Standard before surgery for bulky/pelvic nodes
Netherlands Cancer Institute early vs delayed	Early DSNB vs dissection after surveillance relapse	3-y survival 84% vs 35%; ENE near-universal if delayed	Stage high-risk cN0 groins early

84 Inguinal Node Dissection

1 · Why the groin matters in penile cancer

- Penile SCC has a **locoregional phase**: the groin is a **curable station**.
- **Nodal status is the strongest prognostic factor**; extranodal extension (ENE), even in one node, = pN3.
- Early vs delayed inguinal lymph node dissection (ILND) for impalpable T2–3 nodes: ENE ≈20% vs ≈95%; 3-year disease-specific survival ≈84% vs ≈35%. → Manage groins within **3 months** of diagnosis.
- Adequacy: ≥**8 nodes per side**. Omission predicts mortality.

2 · Indications and risk groups

	Low risk	Intermediate	High risk
NCCN	pTis, pTa, pT1a	pT1b G1–2	≥pT2, or G3/G4
EAU-ASCO	pTa, pTis, G1 pT1a	G2 pT1a	≥pT1b
cN0 — NCCN	Surveillance	Modified/radical bilateral ILND or DSNB	Same
cN0 — EAU-ASCO	Surveillance	Surveillance (compliant) vs DSNB	DSNB; ILND if unavailable

DSNB = dynamic sentinel node biopsy. **Trap**: pT1b G2 is intermediate (NCCN) but high (EAU-ASCO).

- Impalpable high-risk groin: **NCCN leads with ILND** (DSNB if experienced); **EAU-ASCO leads with DSNB** (ILND if unavailable or preferred). Low risk: surveillance, no staging imaging.
- Select on **primary pathology only** (T, grade, LVI/PNI, sarcomatoid), not HPV or nomograms; expert pathology review.
- **cN+**: ≈half metastatic. FNAC → if negative, excisional biopsy → both negative: surveillance; any positive → **bilateral radical ILND**. No antibiotic trial.
- Unilateral palpable nodes → contralateral micrometastasis ≈**30%** → stage the other groin.
- **Bulky/fixed (cN3)**: upfront cisplatin–taxane chemotherapy, then surgery; upfront resection only if chemotherapy-ineligible or vessel invasion imminent. Skin involved → wide excision + gracilis/rectus myocutaneous flap.

3 · Staging work-up

- Indeterminate examination → cross-sectional imaging. Palpable/high risk → ^{18}FDG-PET/CT or CT chest–abdomen–pelvis. Imaging misses deposits <1 cm → no role in the impalpable groin.
- **FNAC**: specific, insensitive → positive = ILND; negative = DSNB or ILND, never discharge.
- **DSNB**: US ± FNAC first (a **tumour-replaced node** reroutes tracer to a neo-sentinel node → false negative) → ^{99m}Tc nanocolloid + blue dye. Glans lesions may drain directly to deep nodes (false negative). Serial sectioning. Positive → ILND †; negative → surveillance.

> **KEY POINT**
> Every groin test fails toward **false negatives**; a negative result never ends the pathway.

4 · Lymphatic anatomy

- P2 Importance and management of Inguinal lymph nodes in CA penis. — *superficial/deep systems, bilateral stepwise no-skip drainage, Daseler's five zones (superomedial = sentinel), superficial 4–25 and deep 1–3 nodes, Cloquet*
- Crossover to contralateral pelvis extremely rare. Cloquet node lies between femoral vein and lacunar ligament.
- **Femoral triangle**: inguinal ligament, **adductor longus** medially, **sartorius** laterally.
- Flaps deep to **Camper's fascia** (capillary network): too thin → necrosis; too thick → nodes left. Incise along Langer's lines.
- Fossa ovalis, lateral → medial: **nerve, artery, vein → never dissect lateral to the femoral artery**.

5 · Templates

Template	Extent	Features
Radical	Whole femoral triangle, superficial + deep	Fascia lata opened; saphenous ligated; sartorius transposed
Modified (Catalona)	Excludes lateral to femoral artery and caudal to fossa ovalis	Shorter incision, saphenous preserved, thicker flaps, no transposition
Superficial	Above fascia lata only	Fascia lata preserved

Modified ILND → frozen section; **positive → convert to radical** †. Negative superficial nodes allow omitting deep dissection.

6 · Open radical ILND

> **ALGORITHM**
> Frog-leg → incision ≈2 cm below and parallel to inguinal ligament → flaps 3 cm above ligament to apex → divide saphenous at apex (spare if minimal disease) → open fascia lata over sartorius and adductor longus → dissect cephalad on femoral vessels → divide saphenous at junction → clear deep nodes to femoral canal → **sartorius transposition** → close femoral canal (inguinal to Cooper's ligament) → suction drains, tack flaps.

- Vessel sealer preferred to monopolar; fibrin sealant no benefit.

7 · Pelvic lymph node dissection (PLND)

- Positive inguinal frozen section → PLND same sitting; otherwise staged.
- † EAU: ≥2 positive inguinal nodes or ENE (pN2–3) → ipsilateral PLND, open or minimally invasive.

8 · Video-endoscopic ILND (VEIL)

- P4 Videoendoscopic inguinal lymphadenectomy — *same template beneath Scarpa's, 15 mm Hg, frozen section gates deep dissection; similar yield, fewer wound complications; non-bulky nodes only*
- Ports ≈25 cm below the ligament; check haemostasis at 5 mm Hg; fewer severe oedema; commonest robotic complication lymphocele.

9 · Complications and post-op care

Complication	Cause	Prevention / management
Wound infection, breakdown (commonest)	Skin flora, dead space	Asepsis, tack flaps, incisional negative pressure
Flap necrosis	Breach of Camper's capillary network	Correct plane, Langer-line incision, modified template, VEIL; debride + flap/negative pressure †
Lymphocele / seroma	Unsealed lymphatics	Sealing devices; drain until **<30 mL/day**; drain symptomatic/infected collections †
Lymphoedema	Lymphatic interruption	Saphenous sparing, modified template, compression hose ≥1 month
Others	Phlebitis, neurapraxia, PE, femoral bleeding	Thromboprophylaxis; stay medial to artery

Early mobilisation. NCCN advises antibiotic prophylaxis (oral cephalosporin days–weeks); authors do not routinely. Root causes: disrupted **skin blood supply** and **lymphatics**; high-volume surgeons reduce both.

10 · Follow-up

Median recurrence ≈1 year; most within 2–3 years. EAU-ASCO: examine 3-monthly × 2 years, 6-monthly to 5 years; image only if node-positive. NCCN images all (CT abdomen–pelvis + chest X-ray), even pN0. Closer after DSNB; self-examination after 5 years.

83 Tumours of the Urethra

1 · Benign urethral tumours

Lesion	Who / where	Clue	Treatment
Leiomyoma	Reproductive-age women; posterior wall	Firm suburothelial mass; grows in pregnancy	Excision; may recur
Haemangioma	Men, any age	Bluish sessile lesion, painless bleeding; posterior → haematospermia; Klippel–Trénaunay–Weber	Fulguration/laser; open excision if extensive
Fibroepithelial polyp	Boys; bulbar/posterior	Polyp on a stalk; filling defect	Transurethral resection

2 · Anatomy → histology → lymphatics

Feature	Male (~20 cm)	Female (~4 cm)
Segments	Anterior: fossa navicularis, pendulous, bulbar. Posterior: membranous, prostatic	Anterior = distal ⅓; posterior = proximal ⅔
Epithelium	Prostatic: urothelium → anterior: columnar → fossa navicularis: squamous	Proximal ⅓ urothelium; distal ⅔ stratified squamous
Glands	Cowper's, Littré	Skene's (prostate homologue)
Histology	Urothelial carcinoma (UC) 50–80%, squamous cell carcinoma (SCC) 10–30%, adenocarcinoma 5–10%. **Anterior: up to 80% SCC; posterior: mostly UC**; adenocarcinoma relatively more in bulbar	UC 28–45%, SCC 19–29%, **adenocarcinoma 28–38%** (near-equal)
Lymphatics (both sexes)	Anterior → **inguinal** (bulbar also external iliac); posterior → **pelvic**	

KEY POINT

Site predicts histology (distal SCC, proximal UC), histology predicts behaviour, and the same anterior–posterior line fixes the nodal basin. In women it is also the **continence boundary**: the distal third can be excised without incontinence.

- Grading: UC low/high grade; SCC/adenocarcinoma G1–G3.

3 · Staging — AJCC 8th edition

T	Penile urethra and female urethra	Prostatic urethra
Ta / Tis	Non-invasive papillary / carcinoma in situ (CIS)	Tis also includes prostatic duct involvement **without stromal invasion**
T1	Subepithelial connective tissue	Subepithelial connective tissue
T2	Corpus spongiosum or periurethral muscle	Prostatic stroma
T3	Corpus cavernosum or anterior vagina	Periprostatic fat
T4	Other adjacent organs (bladder)	Bladder, rectum
N	N1 single, N2 multiple regional nodes (inguinal, pelvic, presacral) — **number, not size**	

- **Evaluation**: EUA + cystourethroscopy, bimanual palpation, biopsy; cytology insensitive. **MRI** for local extent; CT chest if invasive.
- **Palpable inguinal nodes (20–30%) are almost always metastatic** — unlike penile cancer, no antibiotic trial.

4 · Male primary urethral cancer (PUC)

- >60 years. **Urethral stricture** is the commonest risk factor (up to 50%; latency 20–30 years). Also lichen sclerosus, STI, pelvic RT, HPV 16.
- † Periurethral abscess/fistula may present. † Biopsy a stricture that bleeds or recurs rapidly.
- **Prognosis**: stage and location — distal 2–3-fold better; bulbar survival as low as 20–30%.

Pendulous / fossa navicularis (distal)

- Surgery alone controls low-stage disease: transurethral resection → **distal urethrectomy ± partial penectomy** → total urethrectomy + perineal urethrostomy ± total penectomy.
- **Penile preservation** (EAU, NCCN): topical 5-FU for distal CIS, distal urethrectomy + two-stage urethroplasty, glansectomy + graft, anterior urethrectomy + perineal urethrostomy. Positive margin → re-excision or adjuvant RT.
- **Margin: 5 mm** now standard (1 cm traditional). Invasion into proximal penile urethra → total penectomy. Distal CIS can be upstaged.
- **Primary RT**: only early anterior lesions refusing surgery; stricture, ulceration, oedema. Ineffective for bulbar PUC.

Bulbar (proximal anterior)

- Low-stage (few): transurethral resection or segmental excision + anastomosis.
- **Advanced**: radical cystoprostatectomy + pelvic lymphadenectomy + total urethrectomy + penectomy ± inferior pubic rami and urogenital diaphragm. Surgery alone → ~¼ 5-year survival → **multimodal therapy**.

Inguinal nodes (both sexes)

- **No proven benefit of prophylactic inguinal dissection in cN0**. Irradiated primary + cN0 → consider prophylactic groin RT.
- cN1 → neoadjuvant chemotherapy/chemoradiotherapy then lymphadenectomy.

5 · Male posterior (prostatic) urethral cancer

- Usually **secondary to bladder UC**: 20–40% at radical cystectomy (RC); up to 39% of high-risk NMIBC recurs in prostate after intravesical therapy → always evaluate bladder and upper tracts.
- Diagnose by TURP biopsy; often CIS. BCG for high-grade superficial disease: ≥70% response; extensive TURP + BCG ~half complete response.

Involvement	Treatment	5-year survival
Mucosal	TURP + BCG	67–74%
Ductal / acinar (still Tis)	TURP + BCG, or RC ± urethrectomy	67–74%
Stromal invasion	**RC + urethrectomy + pelvic lymphadenectomy** ± neoadjuvant chemotherapy	~36%; follows bladder stage

KEY POINT
The only question is "stroma breached?" TURP before BCG debulks, improves contact and stages. Ductal disease is not T2 — do not jump to cystectomy.

6 · Female urethral cancer

- ~⅓ of PUC; >60 years. Risk: chronic irritation, infection, leukoplakia, caruncle, HPV, **urethral diverticulum (6–10% harbour malignancy**; mostly adenocarcinoma).
- Distal → SCC; proximal → UC or adenocarcinoma. Adenocarcinoma: **mucinous** (commoner; CEA) and **clear cell** (diverticula; may secrete PSA); likely Skene's origin.
- **Location decides continence-sparing excision**; ⅓ distal. MRI pelvis for staging.
- 30–50% locally advanced; ~20% palpable inguinal nodes. **Stage dominates prognosis**; distal > proximal > pan-urethral. Melanoma uniformly poor.

Treatment

Situation	Management
Small non-invasive	Endoscopic resection / laser ± BCG
Low-stage distal	**Distal urethrectomy** (transvaginal) **or RT** (brachytherapy ± external beam, ~60 Gy) — equal; ≥70% cured. Stress incontinence (>40%), meatal stenosis; ~1 in 5 recur locally. RT complications 20–50% (stricture, necrosis, fistula)
Proximal / pan-urethral	**Anterior exenteration**: RC + total urethrectomy + extended pelvic lymphadenectomy + anterior vaginal wall ± hysterectomy, vulvectomy, pubic symphysis/rami. Alone → most die, ~⅓ pelvic recurrence → **multimodal**
Palpable groin nodes	Chemoradiotherapy, chemotherapy or lymphadenectomy; often curable. Pelvic nodes → poor

- RT local control depends on **length of urethra involved**, not histology or size; external beam + brachytherapy beats external beam alone.

7 · Multimodal therapy (both sexes)

- Advanced/proximal disease → **cisplatin-based neoadjuvant chemotherapy** (EAU, NCCN) or **chemoradiotherapy**, then consolidative surgery. Response predicts survival.
- **Chemoradiotherapy**: 45–55 Gy to genitalia, perineum, groins + 5-FU and mitomycin C — preferred for advanced SCC; every responder develops stricture; ~⅓ recur locally. Not for low-stage distal disease.

Histology	Regimen
UC	MVAC; gemcitabine–cisplatin
UC or SCC	Paclitaxel, ifosfamide, cisplatin (TIP); cisplatin, gemcitabine, ifosfamide
Adenocarcinoma	Gemcitabine, 5-FU, leucovorin, cisplatin

Stage (NCCN, anterior male / female)	First line
Tis, Ta, T1	Repeat transurethral resection ± intraurethral BCG/chemotherapy
Male T2 pendulous	Partial urethrectomy ± partial penectomy; positive margin → chemoradiotherapy/surgery/RT

Stage (NCCN, anterior male / female)	First line
Male T2 bulbar	Urethrectomy ± RC; ≥pT3/N1 → adjuvant chemotherapy or chemoradiotherapy
Female T2	RC + urethrectomy, or distal urethrectomy by location, or chemoradiotherapy
≥T3 N0	Chemoradiotherapy ± surgery; or neoadjuvant chemotherapy → surgery/RT
N1	Chemoradiotherapy (preferred for SCC) or systemic therapy ± consolidative surgery

- Recurrence after resection of invasive disease ~50% → urethral, inguinal and systemic surveillance. PD-L1-positive may respond to checkpoint inhibitors.

8 · Urethral recurrence (UR) after radical cystectomy

Male

- Risk **1–15%** (~5%); most within 2 years, ~11% after 5 years.
- **Risk factors**: **prostatic urethral involvement** (15% vs 5%), **positive urethral margin**, cutaneous diversion, CIS, multifocality, prior NMIBC. Orthotopic neobladder (ONB) protective (~5% vs 9%).
- Routine prophylactic urethrectomy abandoned (64–100% of specimens negative).
- **Before ONB**: negative intraoperative frozen section of urethral margin (better than pre-op prostatic biopsy).
- **Urethrectomy** — absolute: positive margin or gross urethral tumour; relative: diffuse CIS, prostatic urethral involvement, multifocal, bladder neck or upper tract tumours. Delayed urethrectomy as good as simultaneous.

Female

- Synchronous urethral involvement 8–13%; main risk **bladder neck involvement** — only 30–50% of those have urethral tumour → not an absolute contraindication.
- ONB if no bladder neck, urethral or anterior vaginal wall involvement and negative frozen section. UR after ONB <5%; most from skip lesions despite negative margins → keep surveillance.

Surveillance and treatment

- Urethral wash cytology (barbotage after cutaneous diversion) for 2 years — no proven survival benefit. Positive → urethroscopy + biopsy.
- **Total urethrectomy is treatment of choice** (including meatus); after ONB → convert to cutaneous diversion. Non-invasive: endoscopic resection (high second-UR risk); **intraurethral BCG for CIS** (not papillary/invasive).
- Invasive UR: median survival ~17 months.

85 Benign Disorders of the Penis and Urethra

1 · Reconstructive surgical technique

- **Cautery**: **bipolar on the urethra** (minimal spongiosal damage); monopolar on subcutaneous tissue and muscle.
- **Bougie à boule** = calibration. **Goodwin sounds** = dilation, safer than van Buren (guidewire lumen).
- **Sutures**: always **absorbable** (permanent = stone nidus); monofilament PDS preferred. Support needed **30–60 days**; chromic lasts 14 days → avoid. **5-0** for most urethral work; **6-0** for single-layer mucosal anastomosis (non-transecting EPA); RB-1 taper needle.
- Limit high lithotomy (neuropathy, compartment syndrome); low lithotomy for graft harvest.

2 · Tissue transfer: grafts and flaps

- P1 Enlist various grafts used in urology and role of BMG in urology — *graft vs flap; take by imbibition then inosculation (~48 h each), immobilise 5–10 days; STSG vs FTSG; other grafts; BMG properties, buccal = lingual*
- Imbibition: graft below core temperature; inosculation: rises to core. Anything impairing bed vascularity impairs take.
- Skin in contact with urine must be **hairless** (pre-harvest laser/electrolysis).
- **Extragenital FTSG** fails in the urethra (greater mass → ischaemic take). Exception: **posterior auricular (Wolfe) graft** — thin, on perforators, behaves like an STSG.
- OMG: **panlaminar plexus**, sacculates less than skin. BMG: more early swelling, numbness; lingual: speech difficulty; labial: worst morbidity. Buccal onlay failures occur in the **first 12 months**.

> **KEY POINT**
> Graft trade-off: dense fine plexus and low mass favour take; carried dermis/lamina gives durability and resists contraction. This explains STSG vs FTSG, failure of extragenital FTSG, the Wolfe exception, and why OMG is ideal.

Flaps (independent axes)

- **Composition**: fasciocutaneous (distal penile circular), fascial (dartos), musculocutaneous (rectus), muscle (gracilis), visceral (omentum).
- **Blood supply**: axial (named vessel — gracilis, penile circular flap) vs random (U-flap, **Z-plasty**). Axial: peninsula, island (skin divided, pedicle intact), free.
- **Movement**: advancement (Y-V), rotation (Martius), transposition (Z-plasty). **Location**: local, distant pedicled, free (radial forearm phalloplasty).

3 · Surgical anatomy

- Deep → superficial: **tunica albuginea → Buck's → dartos → skin**. Buck's encloses deep dorsal vein, dorsal arteries and nerves, splits around spongiosum; dartos is fat-free, carries skin vessels, fuses with Colles.
- Urethra **eccentric (dorsal) in the bulb**, central in shaft; membranous urethra within striated external sphincter.
- **Skin supply**: superficial external pudendal → subdermal plexus (McAninch circular flap); posterior scrotal artery → ventral longitudinal flap.

- **Deep supply**: internal pudendal → common penile → **bulbourethral, dorsal (circumflex branches, glans), cavernosal**.

Situation	Spread
Buck's intact (penile fracture)	Confined to shaft → **eggplant deformity**
Buck's ruptured (straddle)	Within Colles–Scarpa: scrotum, perineum, abdominal wall; limited on thigh by fascia lata
Fournier gangrene	Perineum/scrotum → abdominal wall via Scarpa

> **KEY POINT**
> Two envelopes, two vascular worlds: dartos carries skin supply (skin flaps raised on dartos); Buck's encloses erectile bodies fed by the common penile artery.

4 · Lichen sclerosus (LS)

- P1 Lichen sclerosus (BXO, balanitis xerotica obliterans) — *meatus sentinel, Littre-gland stricture mechanism, spares proximal urethra, biopsy features, RUG first, clobetasol, oral mucosa never skin, perineal urethrostomy*
- Malignant transformation up to **8%** (mostly SCC).
- **Steroid**: clobetasol propionate or betamethasone 0.05% once–twice daily 1–2 months, then alternate days 3–4 months.
- Meatal stenosis: extended meatotomy with inverted-V relaxing incision (prevents puckering) or BMG onlay; restenosis commoner than non-LS. Early meatal/fossa repair may avert panurethral stricture.
- Panurethral: perineal urethrostomy or staged reconstruction (stricture first, skin later).

> **KEY POINT**
> LS is a progressive skin disease sparing the proximal urethra: hence meatus as sentinel, RUG before meatotomy, oral mucosa not skin, high recurrence, and easy perineal urethrostomy.

5 · Other urethral lesions

Lesion	Key features	Management
Urethral haemangioma	Bloody discharge; meatus common; extends beyond view → **MRI**; benign	Observe; small → laser or triamcinolone sclerosant; large → excision + staged repair (avoid tubed grafts)
Reactive arthritis (Reiter)	Arthritis, conjunctivitis, urethritis after Chlamydia/Shigella/Salmonella; HLA-B27 ~2/3; **circinate balanitis**	Urethritis self-limiting
Amyloidosis	Urethral mass, obstruction, haematuria; biopsy	Observe; dilate or urethroplasty if obstructive
Urethrocutaneous fistula	Mostly iatrogenic (hypospadias, phalloplasty); piercing, infection behind stricture	Treat the **underlying process** as well as the tract
Male urethral diverticulum	Congenital (all layers) vs acquired (no smooth muscle; behind stricture, catheter, trauma); dribbling, UTI, stone	Cystourethroscopy + RUG; repair + treat cause

Lesion	Key features	Management
Enlarged utricle	Müllerian remnant, proximal hypospadias	Catheter may enter utricle; UTI, post-void dribble

6 · Prepuce and meatus

- **Paraphimosis (emergency)**: trapped foreskin → oedema → glans necrosis; classically after catheterisation. Reduce by steady compression, thumbs push glans, fingers pull foreskin (granulated sugar aid). Irreducible → emergency **dorsal slit**/circumcision; elective circumcision for recurrence.
- **Phimosis**: physiological in infants (resolves 1–8 years); adults → balanitis, cancer. Steroid, dorsal slit, circumcision.
- **Balanitis**: **Candida** commonest; also STI, chemical, Zoon. Hygiene, antifungal, **check blood glucose**. Recurrent balanitis raises cancer risk.
- P2 Management of Urethral meatal stenosis. — *adult causes, RUG first, ventral meatotomy with 3 sutures, steroid self-dilation, urethroplasty if fossa involved*

Adult circumcision (sleeve)

> **ALGORITHM**
> Penile block → foreskin retracted: mark cuff incision across frenular base → incise to Buck's → foreskin advanced: mark coronal incision ± frenuloplasty → excise sleeve with **bipolar** (adult vessels do not seal by pressure) → quadrant dartos sutures → skin closure

- **HIV**: foreskin rich in CD4+ cells, microtears; circumcision ↓ female-to-male HIV ~**60%**; also ↓ HSV, HPV. MSM: benefit mainly insertive partners.

7 · Penile curvature

- P1 Congenital Penile curvature.. — *chordee without hypospadias vs congenital curvature (young man, large erection); never divide a urethra reaching the glans tip*

Devine–Horton type	Abnormal tissue
I	Spongiosum deficient (chordee without hypospadias, I–III)
II	Dysgenetic Buck's + dartos; urethra/spongiosum normal
III	Dartos only
IV	All normal; asymmetric tunica = **congenital curvature**
V	Congenital short urethra — existence doubted

> **ALGORITHM**
> Excise ventral dysgenetic tissue + mobilise spongiosum glans → perineum → artificial erection → ventral midline septotomy → still curved: dorsal tunical ellipse excision/plication under elevated neurovascular bundle (dorsal arteries may be dominant glans supply)

- **Acquired non-Peyronie** *→ Ch 76*: deep urethrotomy into tunica, neglected fracture, **subclinical fracture** (buckling → lateral scar, indentation), septal tear (mimics PD). Normal erections, no shortening. Treat by **scar excision + graft**, not contralateral plication.

8 · Penile transplantation

Composite allotransplant: natural appearance, erection, voiding — but lifelong immunosuppression, ethical issues. Alternative: phalloplasty.

9 · Adult acquired buried penis (AABP)

- Risks: **obesity**, aggressive circumcision, scrotal lymphoedema, LS.
- **Vicious cycle**: escutcheon → tethered penis → trapped urine → inflammation, LS, scarring → more trapping; **weight loss alone rarely cures**.
- **Pariser** classification: I skin flap, II graft, III scrotoplasty, IV escutcheonectomy, V panniculectomy.
- LS cicatrix alone → trial topical steroid. CT if hernia suspected. **Evaluate urethra**: severe stricture → urethroplasty **before** grafting. Optimise diabetes.

ALGORITHM

Slit phimotic ring → glans traction → circumcising incision, excise non-viable skin → escutcheonectomy (harvest graft from it; keep 2–3 cm skin at base) → tack Scarpa to rectus fascia/pubis → drains → STSG (0.012–0.020 inch), fibrin glue, bolster 5–7 days

- Dressing off ~day 5; drains out <30 mL/day.
- **Scrotal lymphoedema**: congenital (Milroy — infancy; Meige — puberty; tarda — adult), infective (**Wuchereria bancrofti**, LGV), hidradenitis, malignancy, node dissection, radiation. Resect **all** involved tissue; cover with spared perineal/lateral scrotal skin.
- **Foreign body injection** (silicone, paraffin): fibrosis, necrosis, abscess, Fournier. Excise all + STSG; screen for **body dysmorphic disorder**.

10 · Landmark trials

Trial	Design	Result	Take-home
African circumcision trials (South Africa, Kenya, Uganda)	Three RCTs, adult male circumcision vs none	~60% ↓ female-to-male HIV acquisition	Circumcision is an HIV-prevention strategy

86 Urethral Stricture Disease

1 · Definitions, anatomy, aetiology

- P2 P3 Urethral stricture disease: etiology, presentation and management (including recent advances); Urethral stricture: classification, DVIU and surgery after failed VIU — *anterior-only definition, spongiofibrosis, area ∝ diameter², Jordan spectrum, dorsal eccentric bulb, dual supply, aetiology table (idiopathic ≈ iatrogenic), LS via Littre glands*

Site	Correct term
Anterior urethra	Stricture (epithelial scar ± spongiofibrosis)
Membranous urethra after pelvic fracture	Distraction defect; pelvic fracture urethral injury (PFUI)
Posterior urethra after trauma/radical prostatectomy	Posterior urethral stenosis
Prostatic urethra, bladder neck	Contracture or stenosis

- Corpus spongiosum supplied by common penile artery; dorsal 12 o'clock urethrotomy is dangerous in the bulb.
- Congenital stricture: infants before ambulation. LS: microabscesses in Littre glands; antibiotics relieve symptoms, not stricture.

> **KEY POINT**
> Retrograde glanular flow allows the bulb to be transected and mobilised. Where it is lost (hypospadias, common penile artery injury), mobilisation becomes ischaemia — linking vessel-sparing excision and PFUI repair failure.

> **KEY POINT**
> Aetiology predicts biology: iatrogenic and straddle strictures are focal ischaemic/crush scars suited to excision; LS is a progressive skin disease that recurs in any skin used to repair it.

2 · Presentation and evaluation

- **Acute retention**: failed catheter → RUG → **suprapubic cystostomy** rather than blind dilation; filiforms condemned; flexible endoscopy + guidewire under vision.
- Four determinants: **location, length, depth, density**.
- Dynamic RUG + VCUG (McCallum criteria) by treating surgeon; sonourethrography may underestimate spongiofibrosis; flexible urethroscopy — do not dilate at first evaluation; MRI for tumours/PFUI only.
- **Hydraulic trap**: voiding pressure holds proximal diseased urethra open; coning down = scarred.
- **Urethral rest**: suprapubic tube **6–8 weeks** → true length. Calibrate whole urethra at surgery (bougie à boule).

3 · Decision making

- Decide **palliative vs curative** first; periodic dilation acceptable if agreed. **Reconstructive ladder archaic**: go to the procedure most likely to cure.

ALGORITHM

Short (<1.5 cm) bulbar, no dense spongiofibrosis, first treatment → dilation or internal urethrotomy (once) → recurrence → urethroplasty
Bulbar 1–2 cm (up to 3–5 cm proximally) → excision and primary anastomosis (EPA)
Longer bulbar → oral mucosa graft (OMG) onlay ± augmented anastomosis
Penile → onlay (graft or flap); hostile bed/LS/failed hypospadias → staged OMG
Panurethral/exhausted options/elderly → perineal urethrostomy

4 · Dilation and internal urethrotomy (DVIU)

- P3 Urethral stricture: classification, DVIU and surgery after failed VIU — *secondary-intention race, 10–2 o'clock thin spongiosum, complications, second urethrotomy futile, self-dilation, mitomycin C, laser = cold knife, catheter 3–7 days*
- Dilation cures only an epithelial stricture without spongiofibrosis; **bleeding means tearing**. Safest: **balloon** over endoscopic guidewire. **Optilume** paclitaxel balloon (see trial).
- DVIU worse with luminal narrowing >**74%**; no success in non-bulbar strictures; repetition worsens later urethroplasty. Use saline (extravasation).

Favourable	Unfavourable
Bulbar	Penile, fossa
<1.5 cm	Longer
No dense deep spongiofibrosis	Dense (e.g. straddle)
First procedure	Repeat

- Self-catheterisation tapering over **3–6 months** (6-week catheter no better than 3–7 days); recurs when stopped. UroLume withdrawn.

KEY POINT

No endoscopic adjunct removes spongiofibrosis, so endoscopy cures only the short superficial bulbar stricture; repetition spends the patient's reconstructive capital.

5 · Excision and primary anastomosis (EPA)

- **Gold standard**; recurrence rare, mostly within 6 months. Rules: **excise all fibrosis; wide spatulation; tension-free**.
- Length gained by spongiosal mobilisation, freeing Buck's, intracrural space, detaching bulbospongiosus from perineal body (~1–1.5 cm). **The more proximal, the longer the excisable stricture**.
- Absorbable 3-0/4-0 for anastomosis, 4-0 to 6-0 for grafts/flaps; bipolar cautery.

	Transecting	Vessel-sparing transecting	Non-transecting
Bulbar arteries	Divided	Preserved	Preserved
Steps	Excise, spatulate, anastomose	Divide at membranous–bulbar junction, dorsal spatulation	Dorsal stricturotomy → excise scar from within → close transversely
Use	Most	Proximal bulbar	Proximal

Vessel-sparing protects men with poor retrograde supply (hypospadias, trauma, prior surgery); less ED and AUS cuff erosion theoretical, unproven.

6 · Substitution urethroplasty

- **BMG** = workhorse (one-stage onlay, staged); FTSG (not in LS); meshed STSG staged only; Wolfe graft staged hypospadias. Best in the **bulb** (thick vascular bed).
- Graft position: ventral only with spongioplasty; lateral (thin spongiosum, less bleeding); **dorsal (Barbagli)** spread-fixed on corpora.
- **Augmented anastomosis** often beats pure onlay. **Kulkarni**: one-stage panurethral dorsal onlay via perineum with **penile invagination**.
- **Staged**: stage 1 lay open, graft quilted to dartos (not tunica — cannot be tubed); stage 2 Thiersch tubularisation to ~28 Fr after ≥6 months (12 after STSG). For LS, failed hypospadias.

> **KEY POINT**
> A graft lives on its bed — it succeeds where the bed is thick and vascular (bulb, dorsal corpora) and is staged when the bed, or skin disease, cannot be trusted in one step.

- **Flaps**: hairless genital skin on dartos; **Quartey**: skin is a passenger on a fascial flap. Ventral longitudinal (**Orandi**), dorsal transverse, hockey stick (7–9 cm), **circular island** up to **15 cm**; scrotal island not first choice.
- **Fossa navicularis**: dartos transverse (Jordan) or longitudinal (De Sy) island.
- Grafts = flaps; flaps for irradiated/multiply operated urethra. **Onlay beats tube**; tubed flap beats tubed graft. Very long: flap distally + graft proximally.
- **LS**: skin fails → **oral mucosa only**, often staged.
- **BMG perineal urethrostomy**: inverted-U incision → transect spongiosum → dorsal urethrotomy to healthy mucosa → ventral urethra to skin, calibrate 30 Fr → BMG quilted to corpora completes dorsal wall.

7 · Meatal stenosis

- **Child**: post-circumcision meatitis → ventral meatotomy + parental dilation 3×/day for 7–10 days; early → 0.05% clobetasol.
- **Adult**: fossa usually involved → evaluate as stricture. Ventral meatotomy, three sutures; dorsal Y-V if needed.

8 · Pelvic fracture urethral injury (PFUI)

- P3 Describe early and late management of PFUD injuries — *bulbomembranous site, primary realignment debate, cystogram + RUG, bladder neck not predictive, MRI, delayed perineal repair after 3 months, progressive steps, voiding study 21–28 days*
- ~**10%** of pelvic fractures; "apple off its stem" wrong. Primary realignment linked to delayed repair, more endoscopic failures.
- Normal anterior urethra on RUG predicts successful anastomosis. Timing **3–6 months**. Transpubic unnecessary; **core-through not curative**.

> **ALGORITHM**
> On-table endoscopy (exclude bladder stones) → exaggerated lithotomy → perineal incision → bulbospongiosus divided, bulb mobilised → Haygrove sound via suprapubic tract → excise fibrosis → spatulate to **≥30 Fr** → 8–10 tension-free 4-0 PDS sutures over silicone stent
> Progressive steps only if tension: mobilisation (to penoscrotal junction) → intracrural separation → infrapubectomy → corporal rerouting

- Success high **90s %**. Failure mostly **ischaemic stenosis** of mobilised bulb, not anastomotic. Anticholinergic post-op.

Internal pudendal status	Potency	Action
Intact ≥1 side	Often potent	Reconstruct
Reconstituted only	Never potent	Reconstruct; revascularisation for ED later
Bilateral occlusion	—	**Revascularise first** (ischaemic stenosis risk)

Screen at-risk men (dorsal nerve injury, failed repair, lateral-impact fracture): **duplex US → angiography if abnormal**.

9 · Vesicourethral stenosis and fistula

- Deep 3 and 9 o'clock incisions; radial incision + mitomycin C if recalcitrant. Obliterated → robotic redo or perineal repair (high incontinence → later AUS). Suprapubic tube/diversion always options.
- **Rectourethral fistula**: simple (<2 cm, non-irradiated) → perineal or York-Mason; complex (irradiated) → faecal diversion, transperineal **gracilis interposition ± BMG**; diversion often the endpoint after radiation.

10 · Complications and follow-up

- Recurrence (EPA early; grafts within 12 months, decline with time); ED ≈ circumcision rate, higher with long repairs and straddle, often improves; chordee; positioning neuropathy.
- Follow-up: symptoms, †uroflow and PVR, urethroscopy if suspicious.

11 · Landmark trials

Trial	Design	Result	Take-home
ROBUST III	Phase III RCT, Optilume vs endoscopic control after multiple failed endoscopic treatments	1-year stricture-free 74.6% vs 26.8%	Drug-coated balloon beats repeat endoscopy; role as first-line undefined

87 Surgery of the Scrotum and Seminal Vesicles

1 · Surgical anatomy of the scrotum

- Lax rugated skin → **poor tamponade**, haematomas keep enlarging: meticulous haemostasis.
- Layers, testis outwards: albuginea → tunica vaginalis → internal spermatic fascia → cremaster → external spermatic fascia → dartos → skin. Dartos keeps necrotising fasciitis off the testes.
- **Transdermal absorption** ~**40×** other skin; rich in 5α-reductase.

Territory	Artery	Parent
Scrotum, anterior	Superficial and deep external pudendal	Femoral
Scrotum, posterior	Posterior scrotal	Perineal (internal pudendal)
Testis, primary	Testicular	Aorta
Testis, secondary	Deferential	Superior vesical ← internal iliac
Testis, secondary	Cremasteric	Inferior epigastric ← external iliac

KEY POINT
Three trunks (aorta, internal iliac, external iliac) anastomose in the cord (~10 cm above testis) and at the cauda — so the testicular artery can be ligated in Fowler–Stephens orchidopexy and the superior epididymal artery sacrificed in vasectomy reversal.

- Nerves: genital branch of genitofemoral (injury → lost cremasteric reflex); ilioinguinal (anterior); posterior scrotal (pudendal).

2 · Peri-operative principles and access

- Shave with **razor on the day of surgery**. **Antibiotics** (AUA): single IV cephalosporin or clindamycin **only with risk factors**.
- Pre-incisional 0.5% lidocaine (no adrenaline) reduces pain.
- Benign: scrotal incision (median raphe avascular, or transverse). **Possible malignancy**: inguinal incision, clamp cord before handling testis → frozen section; spare if benign.
- **Always close the dartos**: most haematomas come from inadequate dartos closure.

3 · Scrotal wall: excision, scrotectomy, reconstruction

- **Fournier gangrene**: commonest indication for partial scrotectomy. Fluids + broad-spectrum antibiotics + **urgent debridement** (spreads **2–3 cm/h**); mortality ~**20%**. Re-debride every **24–48 h**. Anorectal extension may need colostomy.
- Scrotal violation in NSGCT → partial scrotectomy of scar.
- >**50–60%** skin lost: no immediate reconstruction; testes in thigh pockets, saline gauze or under VAC.

ALGORITHM
Defect <50%, no tension → primary closure · <50% with tension → local scrotal advancement / secondary intention · >50% → VAC, then stretched scrotal advancement or delayed primary closure → no local tissue → fasciocutaneous thigh flap · meshed STSG last.

KEY POINT

The scrotum works because it is lax; graft contracture defeats that, whereas residual scrotal skin stretches enormously and is the best donor.

4 · Hydrocele

- **Congenital**: patent processus; ~**80%** resolve by 18 months. **Acquired**: production–absorption imbalance; men >40 y.
- Treat for discomfort, cosmesis, malignancy; pyocele urgently. **Colour Doppler US first**: excludes tumour, decides scrotal vs inguinal route.

Technique	Steps	Best for
Lord plication	Small opening; radial plicating sutures; no dissection	Thin-walled, smaller; least haematoma, no drain
Jaboulay (Winkelman)	Excise part of parietal layer, evert remainder behind testis	Thin, floppy sacs
Bottleneck	Trim to 2-cm cuff, tack edges	Excisional
Excision	Excise sac, oversew edges	Thick, large, multiloculated; lowest recurrence

KEY POINT

More tunica excised → lower recurrence but more haematoma; plication trades a little recurrence for a dry field. Recurrence after any successful repair **0–6%**.

- **Aspiration + sclerotherapy** (unfit/surgery-averse): aspiration alone recurs → always add sclerosant (doxycycline, tetracycline, alcohol, polidocanol). Check fluid for sperm.
- With **retreatment** resolution **85–96%**. Poor if multiloculated or >**750 mL**.

5 · Spermatocele and epididymal surgery

- Spermatocele: operate only for symptoms; **counsel on obstructive azoospermia** (corpus/cauda). Dissect to stalk, ligate; unroof if inseparable.
- Epididymal tumour: suspected malignancy → inguinal route; proven benign → scrotal microsurgical excision.
- **Epididymectomy**: **poor for chronic epididymitis** (>6 weeks) → last resort; better with a **palpable lesion**. Ligate efferent tubules at caput and stumps; orchidectomy may follow.

6 · Varicocele repair

- P2 Surgical management of bilateral varicocele — *~15% of men; Palomo, laparoscopic, microsurgical subinguinal (lowest recurrence and hydrocele, gold standard) and embolisation compared; microsurgical steps*
- †Palomo misses cremasteric/external spermatic collaterals; Ivanissevich (inguinal) opens external oblique → more pain; embolisation suits post-surgical recurrence (coil migration, radiation).
- Orchialgia: **microsurgical repair** gives more complete relief.

7 · Vasectomy

- **Counsel**: not immediately sterile (contraception until PVSA); repeat vasectomy <**1%**; pregnancy ~**1 in 2000** even after azoospermic PVSA; chronic pain ~**1–2%**.

- **No routine antibiotics**. Topical cream never sole anaesthetic.
- **Isolation** determines morbidity; **occlusion** determines efficacy.

Feature	Conventional (CV)	Minimally invasive (MIV)	No-scalpel (NSV)
Skin opening	1.5–3 cm	≤ **10 mm**	Puncture <10 mm
Status	Difficult cases	**Recommended**: less pain, bleeding, infection	Original MIV (Li Shunqiang, China, 1974)

- Open on the **straight vas** at/above penoscrotal junction (convoluted vas harder now and at reversal).
- NSV (ring clamp, sharp mosquito) defines isolation only.

AUA-recommended occlusion (failure ≤1%)	Note
Mucosal cautery (MC) + fascial interposition (FI)	FI = internal spermatic fascia over one end
MC alone (no clips/ligatures)	Cautery of lumen 0.5–1.5 cm
Open-ended: abdominal end MC + FI, testicular end open	Less congestive epididymal pain; granuloma linked to better reversal

KEY POINT
Every recommended method rests on **mucosal cautery**; ligatures and clips alone are not core methods. Division required; segment excision optional (**1 cm** adequate).

ALGORITHM
PVSA at **8–16 weeks** (≥10–20 ejaculations) → azoospermia or rare non-motile sperm (RNMS, ≤ **100,000/mL**) → stop contraception · motile sperm → repeat at 6 months → still motile = failure → repeat vasectomy.

- **Early failure**: technical (one vas missed), on PVSA. **Late**: recanalisation after successful PVSA (~1 in 2000).
- Complications: haematoma, infection, **sperm granuloma** (expected, not a complication; NSAIDs; persisting <5% → excise), post-vasectomy pain ~ **2%** (cord block → denervation or reversal).

8 · Orchidectomy and prosthesis

- **Bilateral simple**: hormone-sensitive metastatic prostate cancer; cheaper, less gynaecomastia than ADT; survival benefit unproven.
- **Unilateral**: trauma, non-salvageable torsion; orchialgia only after all conservative measures.
- **Epididymis-sparing** (microscope) → better contour; †subcapsular same aim.
- **Orchialgia → inguinal route** (removes more proximal ilioinguinal, iliohypogastric, genitofemoral nerve); persisting pain → resect genital branch.
- **Prosthesis**: offer every man lacking a testis, before cancer orchidectomy. Port pierced ≤ **5 times**, never post-op. **High scrotal subdartos** pouch: least infection/erosion; trans-scrotal more extrusion.

9 · Testicular torsion and fixation

- †**Acute torsion**: explore without imaging delay if suspicion high → detorse, assess → fix or orchidectomy → **always fix contralateral testis**.
- Intermittent torsion (pain <2 h) and postpubertal retractile testis → elective orchidopexy.

Suture fixation	Dartos pouch
Albuginea to dartos (3-0 absorbable); re-torsion possible even with non-absorbable; transparenchymal sutures → inflammation, abscess, spermatogonia loss (animals)	Subdartos pouch, purse-string at cremaster exit; no transparenchymal suture → **preferred when fertility matters**

10 · Chronic scrotal pain (>3 months)

ALGORITHM

History, exam, duplex US → treat cause · none → exclude extrascrotal cause (imaging abdomen, pelvis, spine, hip) → prostatitis/pelvic floor → treat · non-specific → support, NSAIDs, tricyclics, gabapentin ≥3 months → epididymal → epididymectomy · testis/post-vasectomy → cord block: > **50%** relief → microsurgical denervation (or reversal); <50% → pain clinic, inguinal orchidectomy last · post-hernia → ilioinguinal block > **90%** → neurolysis.

- **Microsurgical denervation (MDSC)**: subinguinal; divide ilioinguinal and genital branch → micro-Doppler → **preserve testicular and cremasteric arteries, lymphatics, vas with vessels**; divide everything else.
- **Trifecta nerve complex**: cremaster, perivasal tissue, posterior periarterial tissue/cord lipoma — all divided.

11 · Complications of scrotal surgery

- **Haematoma** (commonest): drain if large/infected/enlarging. Bleeding after orchidectomy = slipped cord ligature → subinguinal re-ligation; falling haematocrit = **retroperitoneal bleed** → extend into retroperitoneum.
- Iatrogenic epididymal/vasal obstruction → azoospermia.

12 · Seminal vesicles

- Wolffian derivatives, bud ~ **12 weeks**; ~ **60%** of seminal fluid. Relations: bladder front, rectum behind, **neurovascular bundles lateral to tips**, ureter anterolateral.
- **MRI most accurate**; TRUS obstruction if SV > **1.5 cm** or ejaculatory duct > **2.3 mm** — confirm by **aspiration showing sperm**.
- Cysts rare: congenital or acquired (EDO). **Zinner syndrome**: SV cyst + **ipsilateral renal agenesis** + EDO (often ectopic ureter into cyst) — †common mesonephric origin.
- Small → observe/aspirate; **transurethral unroofing** preferred for cysts, EDO, stones; large symptomatic → laparoscopic/robotic excision. Stones/haemospermia → seminal vesiculoscopy + holmium; re-obstruction common.

Benign	Malignant
Cystadenoma, papillary adenoma, hydatid cyst, amyloid	Adenocarcinoma, mixed epithelial–stromal, phyllodes, sarcoma, carcinoid, seminoma, yolk sac

- MRI → TRUS biopsy if inconclusive. †Exclude secondary invasion (prostate, bladder, rectum) — far commoner. Adenocarcinoma: radical excision + lymphadenectomy ± adjuvant.

13 · Surgical approaches to the seminal vesicles

Approach	Use	Main risk
Transvesical transtrigonal	Large benign cyst; ectopic ureter into cyst	Urinary leak, ureter
Perivesical	Children, large cyst ± nephroureterectomy	Ureter, vas
Transperitoneal (laparoscopic/robotic)	Bilateral lesions; dominant — opens neither organ	Access injuries
Perineal	At perineal prostatectomy	Neurovascular bundle
Transcoccygeal	Hostile abdomen/perineum	Rectum

- Injuries: rectum → two-layer repair, colostomy if spillage; ureter → reimplant ± psoas hitch/Boari, stent; neurovascular bundle → stay medial.

★ Putting it together

A · Testicular mass: the first week

ALGORITHM

US → AFP, hCG, LDH **before** orchidectomy → sperm bank → **radical inguinal orchidectomy** (never trans-scrotal) → repeat markers (AFP half-life 5–7 days, hCG 1–1.5 days) → CT abdomen–pelvis + chest (→ *Ch 79, 80*).

B · Germ cell tumour by stage

Stage	Seminoma	NSGCT
CS I	Surveillance (occult spread 10–15%); single-dose carboplatin	Low risk → surveillance. High risk (LVI, embryonal-predominant) → surveillance, BEP ×1 or nerve-sparing RPLND
II	≤3 cm: RT, chemo or RPLND; >3 cm → chemo	Normal markers, small nodes → RPLND; else chemo
Advanced (IGCCCG)	Good: BEP ×3 or EP ×4. Intermediate/poor: BEP ×4 (VIP ×4 if lung concern). Full-dose cisplatin	
Residual mass	>3 cm → FDG-PET (≥6 weeks post-chemotherapy)	Any >1 cm → post-chemotherapy RPLND (teratoma)

C · Penile cancer: primary and groin

ALGORITHM

Primary: PeIN → topical 5-FU/imiquimod, laser, glans resurfacing. Invasive → organ-sparing, glansectomy, partial/total penectomy; margins <1 cm acceptable (→ *Ch 82*).

cN0: low risk (pTis/pTa/pT1a) → surveillance; intermediate/high → DSNB or modified ILND; positive → radical.

cN+: FNA positive → bilateral radical ILND; cN3 → neoadjuvant TIP first; pelvic dissection if ≥2 inguinal nodes or ENE (→ *Ch 84*).

D · Anterior urethral stricture

Stricture	Treatment
Bulbar <1.5 cm, first time	Dilation or DVIU once; recurs → urethroplasty
Bulbar 1–2 cm (3–5 cm proximally)	EPA
Longer bulbar	BMG onlay ± augmented anastomosis
Penile	Onlay graft/flap; staged if hostile bed, LS, failed hypospadias
Panurethral, frail	Perineal urethrostomy
PFUI	Suprapubic catheter → delayed progressive perineal anastomosis

Retention: suprapubic catheter, never blind-dilate (→ *Ch 86*).